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Definitions and Information Pertaining to Electrical Apparatus in Hazardous (Classified) Locations

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Definitions and Information Pertaining to Electrical Apparatus in Hazardous (Classified) Locations

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1 Purpose

This Standard provides definitions and information pertaining to protection techniques, terminology, and the installation of electrical apparatus in hazardous (classified) locations and provides an introduction and basic background to the ISA-SP12, Electrical Safety, series of publications and committee activities. It replaces ISA-S12.1, *Definitions and Information Pertaining to Electrical Instruments in Hazardous Atmospheres*, published in 1991.

This document provides a general review of applicable codes and standards, and it should not be used in lieu of those codes and standards for equipment design, manufacture, installation, maintenance and test criteria.

2 Scope

2.1 This Standard provides general guidance for safe design, installation, and maintenance of electrical apparatus in hazardous (classified) locations using appropriate means to prevent ignition of flammable gases and vapors, flammable liquids, combustible dusts, or ignitable fibers or flyings.

2.2 This Standard covers only locations made hazardous, or potentially hazardous, due to the presence of flammable gases or vapors, flammable liquids, combustible dusts, or ignitable fibers or flyings. The Standard is not necessarily relevant to the hazards posed by pyrophoric materials such as explosives or propellants containing their own oxidizers.

2.3 This Standard is concerned only with design, manufacture, installation, maintenance, and test criteria related to arcs, sparks, or hot surfaces produced by electrical apparatus that may cause ignition of flammable gas or vapor-in-air mixtures, clouds or blankets of combustible dust, or easily ignitable fibers or flyings. Apparatus should also comply with the applicable ordinary location requirements (e.g., ISA-S82.01 and ISA-S82.03).

2.4 This Standard does not cover mechanisms of ignition from external sources, such as static electricity or lightning. Some apparatus may produce static electricity. The materials of construction of parts in such apparatus will be an important consideration for application in hazardous locations. The extra precautions necessary for this are beyond the scope of this Standard.

2.5 This Standard does not consider the effects of installation in corrosive atmospheres and the resulting deleterious conditions to the original design integrity of the apparatus. The additional precautions necessary for these conditions are outside the scope of this document.

2.6 This Standard is not an instruction manual for untrained persons. However, it is intended to provide guidance to those involved with the design, manufacture, installation, and maintenance of apparatus used in hazardous (classified) locations. It is also intended to promote uniformity of practice among those skilled in the art. Nothing contained in this Standard is to be construed as a fixed rule without regard to sound engineering judgment.

2.7 For hazardous location apparatus, atmospheric conditions are generally considered to be

a) an ambient temperature range of -20 °C (-4 °F) to 40 °C (104 °F);

b) an oxygen concentration of not greater than 21 percent by volume;

c) a pressure of 86 kPa (12.5 psia) to 108 kPa (15.7 psia); and

d) a relative humidity of 5 percent to 95 percent.

NOTE — Apparatus specified for atmospheric conditions beyond the average is generally permitted but will be subjected to additional requirements.

2.8 Specialized industries such as, but not limited to, mining and shipping may be regulated by the specific authority having jurisdiction. This Standard does not include specific requirements or the rules and regulations unique to any specific industry.

2.9 Various organizations have developed codes, guides, and standards that have substantial acceptance by industry and governmental bodies. Codes, guides, and standards useful in the design and installation of electrical instruments in hazardous (classified) locations are listed in Annex C. These are not considered to be a part of this Standard except for those specific sections of documents referenced elsewhere in this Standard.

2.10 Due to the purpose of this Standard, an attempt was made to avoid originality in principles whenever possible, but rather to utilize definitions, explanations, etc., from accepted publications. As a result, much of the material, except for minor changes, is directly as published by others. While specific credit is not given for each reference, all references are included in Annex B.

3 Definitions

The list is not intended to be all inclusive. Throughout this Standard, reference is made to areas, spaces, locations, and zones. These terms should be considered interchangeable terms designating a three-dimensional space.

3.1 accessible surface:
a surface to which a flammable or combustible mixture has access.

3.2 adequately ventilated area:
an adequately ventilated area is an area that has a ventilation system (natural or artificial) that, as a minimum, prevents the accumulation of gases or vapors to an explosive level. Most standards and recommended practices recommend preventing levels in excess of 25 percent of the Lower Flammable (Explosive) Limit, LFL (LEL).

NOTE — Adequate ventilation of an area alone is not an effective means for the prevention of dust explosions.

3.3 AEx:
required marking prefix for apparatus meeting one or more types of protection in Section 505-2(c) for installation in accordance with Article 505 of the National Electrical Code, NFPA 70: 1999.

3.4 approved:
acceptable to the authority having jurisdiction.

NOTE 1 — See AUTHORITY HAVING JURISDICTION.

NOTE 2 — In determining the acceptability of installations or procedures, equipment, or material, the AUTHORITY HAVING JURISDICTION may base acceptance on compliance with appropriate standards. In the absence of such standards, said authority may require evidence of proper installation, procedure, or use. The AUTHORITY HAVING JURISDICTION may also refer to the listing or labeling practices of product-testing organizations. These organizations are in a position to determine compliance with appropriate standards for the current production of listed or labeled items.

3.5 arcing device:
an electrical make/break component, that is generally interpreted as capable of producing an arc with energy sufficient to cause ignition of a specific ignitable mixture.

3.6 associated apparatus:
apparatus in which the circuits are not intrinsically safe themselves but affect the energy in the intrinsically safe circuits and are relied upon to maintain intrinsic safety. Associated electrical apparatus may be either

a) electrical apparatus that has an alternative type of protection for use in the appropriate hazardous (classified) location, or

b) electrical apparatus not so protected that shall not be used within a hazardous (classified) location.

See also INTRINSIC SAFETY.

3.7 authority having jurisdiction:
the organization, office, or individual that has the responsibility and authority for approving equipment, installations, or procedures.

NOTE — The term AUTHORITY HAVING JURISDICTION is used in a broad manner since jurisdiction and approval agencies vary, as do their responsibilities. Where public safety is primary, the authority having jurisdiction may be a federal, state/provincial, local, other regional department, or an individual such as an inspector from a labor or health department, electrical inspector, or others having statutory authority. An insurance inspection agency, rating bureau, or other insurance company representative may be the authority having jurisdiction. An owner or his designated agent may also assume the role. At government-owned installations, the commanding officer, departmental official, or designated agent may be the authority having jurisdiction.

3.8 automatic:
self-acting, operating by its own mechanism when actuated by some impersonal influence, as for example, a change in current strength, pressure, temperature, or mechanical configuration.

3.9 bonding:
the permanent joining of metallic parts to form an electrically conductive path that will ensure electrical continuity and the capacity to conduct safely any current likely to be imposed.

3.10 cable entry (cable gland):
a device permitting the introduction of an electric cable into an electrical apparatus.

3.11 certified:
generic term referring to apparatus that has been evaluated by a recognized testing agency and confirmed to be in compliance with the applicable standard(s).

NOTE — Some agencies use the terms approved, listed, or labeled apparatus to indicate compliance with the applicable standard.

3.12 Class I location:
a location in which flammable gases or vapors are or may be present in the air in quantities sufficient to produce explosive or ignitable mixtures. See 3.116, 3.118, & 3.120 for definitions of Class I, Zones 0, 1, & 2.

3.13 Class I, Division 1 location:
a location (1) in which ignitable concentrations of flammable gases or vapors can exist under normal operating conditions; (2) in which ignitable concentrations of such gases or vapors may exist frequently because of repair or maintenance operations or because of leakage; or (3) in which breakdown or faulty operation of equipment or processes might release ignitable concentrations of flammable gases or vapors and might also cause simultaneous failure of electrical equipment that could act as a source of ignition.

3.14 Class I, Division 2 location:
a location (1) in which volatile flammable liquids or flammable gases are handled, processed, or used, but in which the liquids, vapors, or gases will normally be confined within closed containers or closed systems from which they can escape only in case of accidental rupture or breakdown of such containers or systems, or in case of abnormal operation of equipment; or (2) in which ignitable concentrations of gases or vapors are normally prevented by positive mechanical ventilation and might become hazardous through failure or abnormal operation of the ventilating equipment; or (3) that is adjacent to a Class I, Division 1 location and to which ignitable concentrations of gases or vapors might occasionally be communicated unless such communication is prevented by adequate positive-pressure ventilation from a source of clean air and effective safeguards against ventilation failure are provided.

3.15 Class II location:
a location that is hazardous because of the presence of combustible dust.

3.16 Class II, Division 1 location:
a location (1) in which combustible dust is in the air under normal operating conditions in quantities sufficient to produce explosive or ignitable mixtures; or (2) in which mechanical failure or abnormal operation of machinery or equipment might cause such explosive or ignitable mixtures to be produced and might also provide a source of ignition through simultaneous failure of electrical equipment, operation of protection devices, or from other causes; or (3) in which combustible dusts of an electrically conductive nature may be present in hazardous quantities.

3.17 Class II, Division 2 location:
a location in which combustible dust is not normally in the air in quantities sufficient to produce explosive or ignitable mixtures and dust accumulations are normally insufficient to interfere with the normal operation of electrical equipment or other apparatus, but combustible dust may be in suspension in the air as a result of infrequent malfunctioning of handling or processing equipment and where combustible dust accumulations on, in, or in the vicinity of the electrical equipment may be sufficient to interfere with the safe dissipation of heat from electrical equipment or may be ignitable by abnormal operation or failure of electrical equipment.

3.18 Class III location:
a location that is hazardous because of the presence of easily ignitable fibers or flyings but in which such fibers or flyings are not likely to be in suspension in the air in quantities sufficient to produce ignitable mixtures.

3.19 Class III, Division 1 location:
a location in which easily ignitable fibers or materials producing combustible flyings are handled, manufactured, or used.

3.20 Class III, Division 2 location:
a location in which easily ignitable fibers are stored or handled (except in the process of manufacture).

3.21 code of practice:
a term referring to a document that describes basic safety features and methods of protection and recommends, e.g., the selection, installation, inspection, and maintenance procedures that should be followed to ensure the safe use of electrical apparatus.

3.22 continuous dilution:
the technique of supplying a protective gas flow continuously to an enclosure containing an internal potential source of flammable gas or vapor for the purpose of diluting any flammable gas or vapor that could be present to a level below its LFL/LEL. Refer to 5.2.2.

3.23 control drawing:
a drawing or other document provided by the manufacturer of the intrinsically safe or associated apparatus that details the allowed interconnections between the intrinsically safe and associated apparatus.

3.24 degree of protection (IP):
a system of rating standard levels of protection provided by apparatus for the protection of persons against contact with live or moving parts inside the apparatus, as well as the protection provided by apparatus against ingress of solids and/or liquids. This type of protection classification is in addition to (and not an alternative to) the types of protection necessary to ensure protection against ignition in hazardous (classified) locations. Definitions are found in IEC Publication 60529.

NOTE — See also ENCLOSURE TYPE.

3.25 dust, combustible:
any finely divided solid material 420 microns or less in diameter (i.e., material passing a U.S. No. 40 sieve) that presents a fire or explosion hazard when dispersed.

3.26 dust-ignitionproof:
a term used to describe an enclosure that will exclude dust and that, when installed in accordance with the original design intent, will not permit arcs, sparks, or heat otherwise generated or liberated inside the enclosure to cause ignition of exterior accumulations or atmosphere suspensions of a specified dust in the vicinity of the enclosure.

3.27 dust layer, combustible:
any surface accumulation of combustible dust that is thick enough to propagate flame or will degrade and ignite.

3.28 dust-protected enclosure:
a term describing an enclosure in which the ingress of dust is not totally prevented, but dust does not enter in sufficient quantity to interfere with the safe operation of the equipment or accumulate in a position within the enclosure where it is possible to cause an ignition hazard.

3.29 dusttight:
an enclosure so constructed that dust will not enter the enclosing case under specified test conditions.

3.30 EEx:
designation of explosion-protected electrical apparatus complying with harmonized CENELEC Standards.

3.31 electrical apparatus:
items applied as a whole or in part for the utilization of electrical energy. These include, among others, equipment for the generation, transmission, distribution, storage, measurement, regulation, conversion, and consumption of electrical energy and items for telecommunication.

3.32 enclosure type:
a North American system of rating standard levels of protection provided to electrical apparatus by enclosures for 1) the protection of persons against contact with live or moving parts inside the enclosure, 2) the protection provided by the enclosure against ingress of solids and/or liquids, 3) the protection provided by the enclosure against the deleterious effects of corrosion, and 4) the protection provided by the enclosure against damage due to the formation of external ice. This enclosure type is in addition to (and not an alternative to) the types of protection necessary to ensure protection against ignition in hazardous (classified) locations. Definitions are found in UL 50 or NEMA 250.

NOTE — See also DEGREE OF PROTECTION.

3.33 encapsulation:
a type of protection in which the parts that could ignite an explosive atmosphere by either sparking or heating are enclosed in an encapsulant in such a way that this explosive atmosphere cannot be ignited. This type of protection is referred to as “m.”

3.34 energized:
electrically connected to a source of potential difference.

3.35 entity concept:
a concept that allows interconnection of intrinsically safe apparatus to associated apparatus not specifically examined in such a combination. The criteria for interconnection is that the voltage (Vmax) and current (Imax) which intrinsically safe apparatus can receive and remain intrinsically safe, considering faults, must be equal to or greater than the voltage (Voc or Vt) and current (Isc or It) levels which can be delivered by the associated apparatus, considering faults and applicable factors. In addition, the maximum unprotected capacitance (Ci) and inductance (Li) of the intrinsically safe apparatus, including interconnecting wiring, must be equal to or less than the capacitance (Ca) and inductance (La) that can safely be connected to the associated apparatus. If these criteria are met, then the combination may be connected without compromising intrinsic safety. For additional information refer to ISA RP12.6.

3.36 entry, direct:
a method of connection of an electrical apparatus to the external circuits by means of the connecting facilities inside the main enclosure or in a terminal compartment having a free opening to the main enclosure. (IEV 426-04-07)

3.37 entry, indirect:
a method of connection of an electrical apparatus to the electrical circuits by means of a terminal box or a plug and socket connection which is external to the main enclosure. (IEV 426-04-08)

3.38 Ex:
designation of explosion-protected electrical apparatus

3.39 Ex component:
part of electrical apparatus for explosive atmospheres which is not to be used alone in such atmospheres and which requires additional evaluation of any electrical apparatus with which it is used.

3.40 Ex Scheme:
an international system of certification for explosion-protected electrical apparatus administered by the IECEE and described by IECEE-04. The eventual goal of this scheme is that a manufacturer of hazardous location electrical apparatus would be able to obtain a single ‘Ex’ Certificate of Conformity from one Certification Laboratory and provide that product in any participating country without legal or technical obstacle and without the need to get it recertified locally.

3.41 explosionproof:
a term used to describe an enclosure that is capable of withstanding an explosion of a specified gas or vapor that may occur within it and of preventing the ignition of a specified gas or vapor surrounding the enclosure by sparks, flashes, or explosion of the gas or vapor within, and that operates at such an external temperature that a surrounding flammable atmosphere will not be ignited thereby. (NEC)

NOTE — See also FLAMEPROOF ENCLOSURE.

3.42 explosive atmosphere:
a mixture with air, under atmospheric conditions, of flammable substances in the form of gas, vapor, mist, or dust in which, after ignition, combustion spreads throughout the unconsumed mixture.

NOTE — See also HAZARDOUS (CLASSIFIED) LOCATION.

3.43 fault (as applicable to intrinsically safe systems):
a defect or electrical breakdown of any component, spacing, or insulation that alone or in combination with other defects or breakdowns may adversely affect the electrical or thermal characteristics of the intrinsically safe system. If a defect or breakdown leads to defects or breakdowns in other components, the primary and subsequent defects and breakdowns are considered to be a single fault. Certain components may be considered not subject to fault when analyses or tests for intrinsic safety are made. See also PROTECTIVE COMPONENT.

3.44 fibers and flyings:
these are materials not normally in suspension in air; and are of larger particle size than dusts. Fibers and flyings include materials such as cotton linters, sawdust, textile fibers, and other large particles that are usually more a fire hazard than an explosion hazard.

3.45 flameproof:
a type of protection of electrical apparatus in which an enclosure will withstand an internal explosion of a flammable mixture which has penetrated into the interior, without suffering damage and without causing ignition, through any joints or structural openings in the enclosure, of an external explosive atmosphere consisting of one or more of the gases or vapors for which it is designed. This type of protection is referred to as “d.”

NOTE — See also EXPLOSIONPROOF APPARATUS.

3.46 flammable (explosive) limits:
the flammable (explosive) limits of a gas or vapor are the lower (LFL/LEL) and upper (UFL/UEL) flammable (explosive) limit percentages by volume of concentration of gas in a gas-air mixture that will form an ignitable mixture.

NOTE 1 — For additional information refer to NFPA 325 and IEC 60079-20.

NOTE 2 — The term *explosive* as it relates to atmospheres and mixtures is in the process of being replaced with the more technically correct term *flammable* throughout many national and international standards. This document, however, continues to use the term *explosive* in many of the definitions that are repeated here as those definitions have been derived from documents that have yet to be updated.

3.47 flammable liquid:
any liquid having a flash point below 37.8 °C (100 °F) and having a vapor pressure not exceeding 275 kPa (40 psia) at 37.8 °C(100 °F).

NOTE — For additional information, refer to NFPA 325.

3.48 flammable gas or vapor:
a gas or vapor which, when mixed with air in certain proportions, will form an explosive gas atmosphere.

3.49 flash point:
the minimum temperature at which a liquid gives off vapor in sufficient concentration to form an ignitable mixture with air near the surface of the liquid, as specified by test.

NOTE — For additional information, refer to NFPA 325.

3.50 ground:
a conducting connection, whether intentional or accidental, between an electrical circuit or apparatus and the earth, or to some conducting body that serves in place of the earth.

3.51 grounded (earthed):
connected to earth or to some conducting body that serves in place of earth.

3.52 group:
a classification of combustible materials.

NOTE — Refer to Clause 4 for additional information.

3.53 hazardous (classified) location:
a location in which fire or explosion hazards may exist due to an explosive atmosphere of flammable gases or vapors, flammable liquids, combustible dust, or easily ignitable fibers or flyings.

NOTE — See also EXPLOSIVE ATMOSPHERE.

3.54 hermetically sealed device:
a device that is sealed against the entrance of an external atmosphere and in which the seal is made by fusion, e.g., soldering, brazing, welding or the fusion of glass to metal.

3.55 temperature apparatus:
as specified by NEC, Article 501-5(a)(1), the term *high temperature apparatus* is to be interpreted as apparatus in which the maximum operating temperature (including ambient temperature effect) exceeds 80 percent of the autoignition temperature in degrees Celsius (°C) of the gas or vapor involved.

3.56 identified (as applied to equipment):
recognizable as suitable for the specific purpose, function, use, environment, application, etc., where described in a particular requirement, e.g. NEC.

NOTE — Suitability of equipment for a specific purpose, environment, or application may be determined by a qualified testing laboratory, inspection agency, or other organization concerned with product evaluation. Such identification may include labeling or listing. For additional information see *labeled* and *listed*.

3.57 ignition (autoignition) temperature (AIT):
the minimum temperature required to initiate or cause self-sustained combustion of a solid, liquid, or gas independently of the heating or heating elements.

NOTE 1 — For additional information refer to NFPA 325.

NOTE 2 — A distinction is made between ignition temperature and flash point. See FLASH POINT.

3.58 ignition capable:
apparatus or wiring that under normal conditions, or under specified abnormal conditions, can release sufficient electrical or thermal energy to cause ignition of a specific explosive atmosphere.

3.59 increased safety:
a type of protection applied to electrical apparatus that does not produce arcs or sparks in normal service and under specified abnormal conditions, in which additional measures are applied so as to give increased security against the possibility of excessive temperatures and of the occurrence of arcs and sparks. This type of protection is referred to as “e.”

3.60 internal wiring:
wiring and electrical connections that are made within apparatus by the manufacturer. Within racks or panels, interconnections between separate pieces of apparatus made in accordance with detailed instructions from the apparatus manufacturer are also considered to be internal wiring.

3.61 intrinsic safety:
a type of protection in which a portion of the electrical system contains only intrinsically safe apparatus, circuits, and wiring that is incapable of causing ignition in the surrounding atmosphere. No single device or wiring is intrinsically safe by itself (except for battery-operated, self-contained apparatus such as portable pagers, transceivers, gas detectors, etc., which are specifically designed as intrinsically safe self-contained devices) but is intrinsically safe only when employed in a properly designed intrinsically safe system. This type of protection is referred to as "i."

NOTE — See also ASSOCIATED APPARATUS.

3.62 intrinsic safety barrier:
a component containing a network designed to limit the energy (voltage and current) available to the protected circuit in the hazardous (classified) location under specified fault conditions.

3.63 intrinsically safe circuit:
a circuit in which any spark or thermal effect, produced either normally or in specified fault conditions, is incapable, in the specified test conditions, of causing ignition of a given explosive atmosphere.

3.64 intrinsically safe electrical apparatus:
electrical apparatus in which all the circuits are intrinsically safe circuits.

3.65 intrinsic safety ground bus:
a grounding system that has a dedicated conductor separate from the power system so that ground currents will not normally flow and that is reliably connected to a ground electrode.

NOTE — For further information, refer to Article 504 of NEC, or Section 10 of CSA C22.1, or ISA RP 12.6.

3.66 intrinsically safe system:
an assembly of interconnected intrinsically safe apparatus, associated apparatus, other apparatus, and interconnecting cables in which those parts of the system that may be used in hazardous (classified) locations are intrinsically safe circuits.

3.67 labeled:
equipment or materials with a label, symbol, or other identifying mark of an organization that is acceptable to the authority having jurisdiction and concerned with product evaluation that maintains periodic inspection of production of labeled equipment or materials and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner.

NOTE — Some agencies use the term *approved, listed*, or *certified* to indicate compliance with the applicable standard.

3.68 liquid, combustible:
a liquid having a flash point at or above 37.8°C (100°F). Combustible liquids are subdivided as follows:

a) Class II liquids include those having flash points at or above 37.8°C (100°F) and below 60°C (140°F).

b) Class IIIA liquids include those having flash points at or above 60°C (140°F) and below 93°C (200°F).

c) Class IIIB liquids include those having flash points at or above 93°C (200°F).

NOTE — For additional information, refer to NFPA 325. It should also be noted that these classes have no relation to the hazardous location classes.

3.69 listed:
equipment or materials included in a list published by an organization acceptable to the authority having jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of listed equipment or materials, and whose listing states either that the equipment or material meets appropriate designated standards or has been tested and found suitable for use in a specified manner.

NOTE — The means for identifying listed equipment may vary for each organization concerned with product evaluation, some of which do not recognize equipment as listed unless it is also labeled. The authority having jurisdiction should utilize the system employed by the listing organization to identify a listed product.

3.70 lower explosive limit (LEL):
refer to FLAMMABLE (EXPLOSIVE) LIMITS.

3.71 lower flammable limit (LFL):
refer to FLAMMABLE (EXPLOSIVE) LIMITS.

3.72 maintenance, corrective:
any maintenance activity that is not normal in the operation of equipment and requires access to the equipment's interior. Such activities are expected to be performed by a qualified person. Such activities typically include locating causes of faulty performance, replacement of defective components, adjustment of internal controls, and the like.

3.73 maintenance, operational:
any maintenance activity, excluding corrective maintenance, intended to be performed by the operator and required in order for the equipment to serve its intended purpose. Such activities typically include the correcting of *zero* on a panel instrument, changing charts, record keeping, adding ink, and the like.

3.74 make/break component:
components having contacts that can interrupt a circuit (even if the interruption is transient in nature). Examples of make/break components are relays, circuit breakers, servo potentiometers, adjustable resistors, switches, connectors, and motor brushes.

3.75 maximum surface temperature:
the highest temperature attained by a surface accessible to flammable gases, vapors, or combustible dusts under conditions of operation within the ratings of the apparatus (including specified abnormal conditions).

3.76 minimum cloud ignition temperature:
the minimum temperature at which a combustible dust atmosphere will autoignite and propagate an explosion.

3.77 minimum dust layer ignition temperature:
the minimum temperature of a surface that will ignite a dust on it after a long time (theoretically, until infinity). In most dusts, free moisture has been vaporized before ignition.

3.78 minimum explosive (dust) concentration:
the minimum concentration of a dust cloud that, when ignited, will propagate a flame away from the source of ignition.

3.79 minimum ignition energy (MIE):
the smallest amount of energy that can ignite the most easily ignitable mixture of a specific gas or vapor-in-air mixture or dust-in-air mixture.

3.80 maximum experimental safe gap (MESG):
the maximum clearance between two parallel metal surfaces that has been found, under specified test conditions, to prevent an explosion in a test chamber from being propagated to a secondary chamber containing the same gas or vapor at the same concentration.

3.81 minimum igniting current ratio (MIC Ratio):
the ratio derived by dividing the minimum current required from an inductive spark discharge to ignite the most easily ignitable mixture of a gas or vapor by the minimum current required from an inductive spark discharge to ignite methane under the same test conditions.

NOTE — For additional information, refer to IEC 60079-3.

3.82 nonautomatic:
non-self-acting — requiring personal intervention for control. As applied to an electric controller, nonautomatic control does not necessarily imply a manual controller, but only that personal intervention is necessary.

3.83 nonhazardous (unclassified) location:
a location in which fire or explosion hazards are not expected to exist specifically due to the presence of flammable gases or vapors, flammable liquids, combustible dusts, or ignitable fibers or flyings. Such a location may also be referred to as a safe area.

3.84 nonincendive circuit:
a circuit, other than field wiring, in which any arc or thermal effect produced, under intended operating conditions of the equipment, is not capable, under specified test conditions, of igniting the flammable gas-, vapor-, or dust-air mixture. See also 3.87, NONINCENDIVE FIELD WIRING.

3.85 nonincendive component:
a component having contacts for making or breaking an ignition-capable circuit and in which the contacting mechanism is constructed so that the component is incapable of igniting the specified explosive atmosphere. The housing of a nonincendive component is not intended to (1) exclude the flammable atmosphere or (2) contain an explosion. This type of protection is referred to as “nC.”

3.86 nonincendive equipment:
equipment having electrical/electronic circuitry and components that are incapable, under normal conditions, of causing ignition of the flammable gas-, vapor-, or dust-air mixture due to arcing or thermal effect. This type of protection is referred to as “nA,” “nC,” or “nR.”

3.87 nonincendive field wiring:
wiring that enters or leaves an equipment enclosure and, under normal operating conditions of the equipment, is not capable, due to arcing or thermal effects, of igniting the flammable gas-, vapor-, or dust-air mixture. Normal operation includes opening, shorting, or grounding the field wiring. See also 3.84, NONINCENDIVE CIRCUIT.

3.88 normal conditions:
equipment is generally considered to be under normal conditions when it conforms electrically and mechanically with its design specifications and is used within the limits specified by the manufacturer.

3.89 oil-immersion:
type of protection in which the electrical apparatus or parts of the electrical apparatus are immersed in a protective liquid in such a way that an explosive atmosphere which may be above the liquid or outside the enclosure cannot be ignited. This type of protection is referred to as “o.”

3.90 powder filling:
a type of protection in which the parts capable of igniting an explosive atmosphere are fixed in position and completely surrounded by filling material to prevent the ignition of an external explosive atmosphere. This type of protection is referred to as “q.”

NOTE — This type of protection may not prevent the surrounding explosive atmosphere from penetrating into the apparatus and Ex components and being ignited by the circuits. However, due to the small free volumes in the filling material and due to the quenching of a flame that may propagate through the paths in the filling material, an external explosion is prevented.

3.91 pressurization:
the technique of guarding against the ingress of the external atmosphere into an enclosure by maintaining a PROTECTIVE GAS therein at a pressure above that of the external atmosphere. This type of protection is referred to as “p.”

3.92 pressurization, Type X:
a method of reducing the classification within an enclosure from Division 1/Zone 1 to nonhazardous (unclassified). See 5.2.1.

3.93 pressurization, Type Y:
a method of reducing the classification within an enclosure from Division 1/Zone 1 to Division 2/Zone 2. See 5.2.1.

3.94 pressurization, Type Z:
a method of reducing the classification within an enclosure from Division 2/Zone 2 to nonhazardous (unclassified). See 5.2.1.

3.95 protection, type of:
the specific measures applied to electrical apparatus to avoid ignition of a surrounding explosive atmosphere. Examples are “e” and “n.”

3.96 protective component (as applied to intrinsic safety):
a component that is so unlikely to become defective in a manner that will lower the intrinsic safety of the circuit that it may be considered not subject to fault when analyses or tests for intrinsic safety are made.

3.97 protective gas:
the gas used for pressurization or for the dilution of flammable gases to a level well below their lower explosive limit, usually below 25 percent LFL/LEL. The *protective gas* may be air, nitrogen, other nonflammable gas, or a mixture of such gases.

3.98 purging:
in a pressurized enclosure, the operation of passing a quantity of PROTECTIVE GAS through the enclosure and ducts, so that the concentration of the explosive gas atmosphere is brought to a safe level.

3.99 qualified person:
one familiar with the construction and operation of the equipment and the hazards involved.

3.100 restricted breathing:
a protection technique in which the tightness of an enclosure is assured so that short-term presence of a flammable gas or vapor cloud around the enclosure will not cause the concentration inside the enclosure to reach the LFL/LEL because of breathing or diffusion. This type of protection is referred to as “nR.”

3.101 safe area:
refer to NONHAZARDOUS (UNCLASSIFIED) LOCATION.

3.102 seal, cable, explosionproof:
a cable termination fitting filled with compound and designed to contain an explosion in the enclosure to which it is attached or to minimize passage of flammable gases or vapors from one location to another. A conduit seal in combination with a cable termination fitting may also be used as a cable seal.

3.103 seal, conduit, explosionproof:
a sealing fitting, filled with a poured potting compound, designed to contain an explosion in the enclosure to which it is attached and to minimize passage of flammable gases or vapors from one location to another.

3.104 seal, factory:
a construction where components capable of initiating an internal explosion due to arcing, sparking, or thermal effects under normal conditions are isolated from the wiring system by means of factory installed flameproof seal or joint for the purpose of eliminating the need for an external, field-installed conduit seal and, in some cases, a field-installed cable seal.

3.105 sealed device:
a device so constructed that it cannot be opened during normal operational conditions or operational maintenance; it has a free internal volume less than 100 cubic centimeters (6.1 cubic inches) and is sealed to restrict entry of an external atmosphere. This type of protection is referred to as “nC.”

3.106 simple apparatus (as applied to intrinsic safety):
a device that will not generate or store more than 1.2 V, 0.1 A, 25 mW, or 20 μJ. Examples are: switches, thermocouples, light-emitting diodes, and resistance temperature detectors (RTDs).

3.107 source of release:
a point from which flammable gases or vapors, flammable liquid, combustible dusts, or ignitable fibers or flyings may be released into the atmosphere.

3.108 special protection:
a protection technique other than those that have been standardized. This type of protection is referred to by IEC 60079-0 as “s.”

3.109 temperature, ambient:
the temperature of air or other media where electrical apparatus is to be used.

3.110 temperature identification number (T Code/Temperature Class):
a system of classification by which one of 14 temperature identification numbers (for zones, six temperature classes) is allocated to apparatus. The temperature identification number represents the maximum surface temperature of any part of the apparatus that may come in contact with the flammable gas or vapor mixture. See Table 1.

3.111 type of protection:
refer to PROTECTION, TYPE OF.

3.112 upper explosive limit (UEL):
refer to FLAMMABLE (EXPLOSIVE) LIMITS.

3.113 upper flammable limit (UFL):
refer to FLAMMABLE (EXPLOSIVE) LIMITS.

3.114 zone:
a method of specifying the probability that a location is made hazardous by the presence, or potential presence, of flammable concentrations of gases and vapors, or combustible mixtures of dusts.

3.115 Zone 0 (IEC):
an area in which an explosive gas atmosphere is present continuously or for long periods. (IEV 426-03-03, modified).

3.116 Zone 0, Class I (NEC):
a Class I, Zone 0 location is a location (1) in which ignitable concentrations of flammable gases or vapors are present continuously; or (2) in which ignitable concentrations of flammable gases or vapors are present for long periods of time. (NEC Section 505-9(a))

3.117 Zone 1 (IEC):
an area in which an explosive gas atmosphere is likely to occur in normal operation. (IEV 426-03-04)

3.118 Zone 1, Class I (NEC):
a Class I, Zone 1 location is a location (1) in which ignitable concentrations of flammable gases or vapors are likely to exist under normal operating conditions; or (2) in which ignitable concentrations of flammable gases or vapors may exist frequently because of repair or maintenance operations or because of leakage; or (3) in which equipment is operated or processes are carried on, of such a nature that equipment breakdown or faulty operations could result in the release of ignitable concentrations of flammable gases or vapors and also cause simultaneous failure of electrical equipment in a mode to cause the electrical equipment to become a source of ignition; or (4) that is adjacent to a Class I, Zone 0 location from which ignitable concentrations of vapors could be communicated, unless communication is prevented by adequate positive-pressure ventilation from a source of clean air and effective safeguards against ventilation failure are provided. (NEC Article 505-9(b))

3.119 Zone 2 (IEC):
an area in which an explosive gas atmosphere is not likely to occur in normal operation and, if it does occur, is likely to do so only infrequently and will exist for a short period only. (IEV 426-03-05, modified)

3.120 Zone 2, Class I (NEC):
a Class I, Zone 2 location is a location (1) in which ignitable concentrations of flammable gases or vapors are not likely to occur in normal operation, and if they do occur, will exist only for a short period; or (2) in which volatile flammable liquids, flammable gases, or flammable vapors are handled, processed, or used, but in which the liquids, gases, or vapors normally are confined within closed containers or closed systems from which they can escape only as a result of accidental rupture or breakdown of the containers or system, or as the result of the abnormal operation of the equipment with which the liquids or gases are handled, processed, or used; or (3) in which ignitable concentrations of flammable gases or vapors normally are prevented by positive mechanical ventilation, but which may become hazardous as the result of failure or abnormal operation of the ventilation equipment; or (4) that is adjacent to a Class I, Zone 1 location from which ignitable concentrations of flammable gases or vapors could be communicated, unless such communication is prevented by adequate positive-pressure ventilation from a source of clean air, and effective safeguards against ventilation failure are provided. (NEC Section 505-9(c))

3.121 Zone 20 (IEC):
an area in which combustible dust, as a cloud, is present continuously or frequently, during normal operation, in sufficient quantity to be capable of producing an explosible concentration of combustible dust in mixture with air and/or where layers of dust of uncontrollable and excessive thickness can be formed. This can be the case inside dust containment where dust can form explosible mixtures frequently or for long periods of time. This occurs typically inside equipment. (IEC 1241-3 - 2.11)

3.122 Zone 21 (IEC):
an area not classified as Zone 20 in which combustible dust, as a cloud, is likely to occur during normal operation, in sufficient quantity to be capable of producing an explosible concentration of combustible dust in mixture with air. This zone can include, among others, areas in the immediate vicinity of powder filling or

emptying points and areas where dust layers occur and are likely in normal operation to give rise to an explosible concentration of combustible dust in mixture with air. (IEC 1241-3 - 2.12)

3.123 Zone 22 (IEC):
an area not classified as Zone 21 in which combustible dust, as a cloud, can occur infrequently, and persist only for a short period, or in which accumulations or layers of combustible dust can give rise to an explosive concentration of combustible dust in mixture with air. This zone can include, among others, areas in the vicinity of equipment containing dust, and in which dust can escape from leaks and form deposits (e.g. milling rooms in which dust can escape from the mills and then settle). (IEC 1241-3 - 2.13)

4 Area (location) classification

Area classification schemes should specify the kind of flammable material that may be present and the probability that it will be present in ignitable concentrations. Area classification schemes and systems of material classification have been developed to provide a succinct description of the hazard so that appropriate safeguards may be selected. The type of protection technique selected and the level of protection it must provide depend upon the potential hazard caused by using electrical apparatus in a location in which a combustible, flammable, or ignitable substance may be present.

4.1 North American methods

In the United States, the area classification definitions are stated in Articles 500 and 505 of the National Electrical Code, (NEC) NFPA 70. In Canada, similar definitions are given in the Canadian Electrical Code (CEC), Part 1, Section 18 and Annex J18 (CSA C22.1).

Various organizations have developed numerous guides and standards that have substantial acceptance by industry and governmental bodies for area classification. Refer to Annex C.

Area classification definitions used in the United States and Canada include the following:

a) CLASS — the generic form of the flammable materials in the atmosphere, which may include gas or vapor, dusts, or easily ignitable fibers or flyings (see Clause 3 for detailed definitions);

b) DIVISION (or ZONE) —an indication of the probability of the presence of the flammable material in ignitable concentration (see Clause 3 for detailed definitions); and

c) GROUP — the exact nature of the flammable material (see 4.1.1 & 4.1.2).

4.1.1 Groups (NEC Article 500 / CEC Annex J18)

The United States and Canadian Electrical Codes recognize seven groups: Groups A, B, C, D, E, F, and G. Groups A, B, C, and D apply to Class I locations; Groups E, F, and G apply to Class II Locations. In NEC these groups are defined as:

Group A - Acetylene

Group B - Flammable gas, flammable liquid-produced vapor, or combustible liquid-produced vapor mixed with air that may burn or explode, having either a MAXIMUM EXPERIMENTAL SAFE GAP (MESG) less than or equal to 0.45 mm or a MINIMUM IGNITING CURRENT RATIO (MIC RATIO) less than 0.4.

NOTE — A typical Class I, Group B material is hydrogen.

Group C - Flammable gas, flammable liquid-produced vapor, or combustible liquid-produced vapor mixed with air that may burn or explode, having either MAXIMUM EXPERIMENTAL SAFE GAP (MESG) values greater

than 0.45 mm and less than or equal to 0.75 mm or a MINIMUM IGNITING CURRENT RATIO (MIC RATIO) greater than or equal to 0.4 and less than or equal to 0.80.

NOTE — A typical Class I, Group C material is ethylene.

Group D - Flammable gas, flammable liquid-produced vapor, or combustible liquid-produced vapor mixed with air that may burn or explode, having a MAXIMUM EXPERIMENTAL SAFE GAP (MESG) greater than 0.75 mm or a MINIMUM IGNITING CURRENT RATIO (MIC RATIO) greater than 0.80, or gases or vapors of equivalent hazard.

NOTE — A typical Class I, Group D material is propane.

Additional information on group classification can be found in NFPA 497, Classification of Flammable Liquids, Gases or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas.

Group E – Atmospheres containing combustible metal dusts, including aluminum, magnesium, and their commercial alloys, or other combustible dusts whose particle size, abrasiveness, and conductivity present similar hazards in the use of electrical equipment.

Group F – Atmospheres containing combustible carbonaceous dusts that have more than 8 percent total entrapped volatiles (see ASTM D3175-89 for coal and coke dusts) or that have been sensitized by other materials so that they present an explosion hazard. Coal, carbon black, charcoal, and coke dusts are examples of carbonaceous dusts.

Group G – Atmospheres containing other combustible dusts, including flour, grain, wood flour, plastic, and chemicals.

Additional information on Group Classification can be found in NFPA 499 Classification of Combustible Dusts and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas.

4.1.2 Groups (NEC Article 505/CSA C22.1 Section 18/IEC 60079-12/prEN 60079-12)

These groups are defined as:

Group IIC - Flammable gas, flammable liquid-produced vapor, or combustible liquid-produced vapor mixed with air that may burn or explode, having either MAXIMUM EXPERIMENTAL SAFE GAP (MESG) less than or equal to 0.5 mm or MINIMUM IGNITING CURRENT RATIO (MIC RATIO) less than 0.45, or gases or vapors of equivalent hazard.

NOTE 1 — This group is similar to a combination of Groups A & B as described in 4.1.1, although the MESG and MIC RATIO numbers are slightly different.

NOTE 2 — Typical gases include acetylene, carbon disulfide, hydrogen, and gases or vapors of equivalent hazard.

Group IIB - Flammable gas, flammable liquid-produced vapor, or combustible liquid-produced vapor mixed with air that may burn or explode, having either MAXIMUM EXPERIMENTAL SAFE GAP (MESG) values greater than 0.5 mm and less than or equal to 0.9 mm or MINIMUM IGNITING CURRENT RATIO (MIC RATIO) greater than or equal to 0.45 and less than or equal to 0.80, or gases or vapors of equivalent hazard.

NOTE 1 — This group is similar to Group C as described in 4.1.1, although the MESG and MIC RATIO numbers are slightly different.

NOTE 2 — Typical gases include ethylene and gases or vapors of equivalent hazard.

Group IIA - Flammable gas, flammable liquid-produced vapor, or combustible liquid-produced vapor mixed with air that may burn or explode, having MAXIMUM EXPERIMENTAL SAFE GAP(MESG) greater than 0.9 mm or MINIMUM IGNITING CURRENT RATIO (MIC RATIO) greater than 0.80, or gases or vapors of equivalent hazard.

NOTE 1 — This group is similar to Group D as described in 4.1.1, although the MESG number is slightly different.

NOTE 2 — Typical gases include propane and gases or vapors of equivalent hazard.

Additional information on group classification can be found in IEC 60079-12 & IEC 60079-20.

4.2 Additional background information

4.2.1 History

Historically, the topic of Hazardous (Classified) Locations first appeared in the National Electrical Code (NEC) in 1923, when a new article entitled “Extra-Hazardous Locations” was accepted. This article addressed rooms or compartments in which highly flammable gases, liquids, mixtures or other substances were manufactured, used, or stored. In 1931, “Classifications” consisting of Class I, Class II, etc., for the hazardous locations were defined. However it was not until 1935 that Groups were introduced into the NEC. (Note: Divisions were introduced into the NEC in 1947.) The four gas Groups, A, B, C, and D, complemented the design of electrical equipment used in hazardous (classified) locations and were defined based on the level of hazard associated with explosion pressures of specific atmospheres and the likelihood that the effects of that explosion could be transmitted outside the enclosure. Group A was defined as atmospheres containing acetylene. Group B was defined as atmospheres containing hydrogen or gas or vapors of equivalent hazard. Group C was defined as atmospheres containing ethyl ether vapor; and Group D was defined as atmospheres containing gasoline, petroleum, naphtha, alcohols, acetone, lacquers solvent vapors, and natural gas.

Despite the fact that the introduction of these Groups was done without standardized testing and without the advantage of today’s technological advances or equipment, these definitions have changed little since that time. The first major testing, in fact, was only conducted in the late 1950s, when engineers at Underwriters Laboratories developed a test apparatus that provided a means to determine how various materials behaved with respect to explosion pressures and transmission, when the specific combustible material was ignited in the test vessel. This apparatus, called the Westerberg Explosion Test Vessel, provided standardized documentation of a factor called the Maximum Experimental Safe Gap (MESG) and permitted other materials to be “classified by test” into one of the four gas groups. The results of these tests are contained in Underwriter Laboratories (UL) Bulletin Nos. 58 and 58A (reissued in July, 1993, as UL Technical Report No. 58). In 1971, the International Electrotechnical Commission (IEC) published IEC 79-1A defining a different type of apparatus for obtaining MESG results. While the two MESG test apparatus are physically different in both size and shape, the results are statistically comparative, although in some cases differences have been observed. A sample of values is shown in the following table:

Material	Westerberg apparatus MESG in mm	IEC apparatus MESG in mm
Propane	0.92	0.94
Ethylene	0.69	0.65
Butadiene	0.79	0.79
Diethyl ether	0.30(‡ 0.60)	0.87
Hydrogen	0.08 (‡ 0.23)	0.29

‡ Additional testing on the Westerberg Apparatus has demonstrated that this theory was true, and the MESG value for diethyl ether more than doubled. Further, Westerberg apparatus testing has also shown that the hydrogen MESG value is 0.23 mm.

Papers have been written to attempt to explain the reasons for these differences in the test data. One, by H. Phillips, entitled “Differences Between Determinations of Maximum Experimental Safe Gaps in Europe and U.S.A.,” appeared in a 1981 edition of the *Journal of Hazardous Materials* and cited a condition of spontaneous combustion in one portion of the Westerberg Apparatus, which was reflected in materials, like diethyl ether, having low ignition temperatures.

While acetylene remains segregated in Group A because of the high explosion pressure, which results from its very fast flame speed, newer test methodologies have defined other types of protection methods that now consider acetylene and hydrogen to be of equivalent hazard. One such method examines the MINIMUM IGNITION CURRENT required to ignite a specific combustible material. This testing produced more variability when the results of specific combustible materials were compared. However, it was found that the minimum ignition currents of one test could be favorably compared with those of other tests if a ratio value based on methane was applied. This testing has resulted in the generation of MIC Ratio data.

Other testing has been performed when it was incorrectly assumed that factors called minimum ignition energy (MIE) and autoignition temperature (AIT) were related and could be used to place materials into Groups. The fact that these were independent factors resulted in deletion of AITs as a basis for Group determination in the 1971 NEC.

MIEs have been found to exhibit theoretical results, which do not translate into practical designs that can be applied to actual electrical devices with their associated energy levels.

Since the primary concern is to have electrical devices that can safely operate when used in locations classified by Class, Group, and Division, the delineations for the gas groups have been defined on the basis of MESG and MIC RATIO.

Further details may be found in NFPA 497.

4.2.2 Division 2 concept

The concept of Division 2, a location in which flammable material will be present only occasionally, was initiated in North America. It was recognized that if the probability of the presence of flammable material is low, the protective measures necessary to prevent an explosion can be less restrictive (and normally also much less expensive) than those required in Division 1 locations. In Division 1 locations the probability that the flammable material is present is much higher than in Division 2 locations because in the former, the flammable atmosphere is present frequently during normal operations. Although many international corporations, particularly oil and chemical companies, used the North American nomenclature and practice, it was not until the 1960s that Division 2 began to be accepted outside North America. At the present time the concept of Division 2 area classification is recognized universally. The relaxation of protective measures in Division 2 has not yet reached the same level of acceptance, however. In Japan, for example, the less onerous levels of protection permitted in Division 2 by the National Electrical Code and the Canadian Electrical Code are not yet recognized.

4.2.3 Zone concept

In the 1960s Europe made its own contribution to the practice of area classification by introducing the concept of Zone 0. The intent of defining Zone 0 was to define those locations in which the flammable material is present such a high percentage of the time that extraordinary measures should be taken to protect against ignition by electrical apparatus. The objective of defining Zone 0 and Zone 1 was to allow a less restrictive practice in the remainder of locations formerly classified within Division 1. IEC has recognized three levels of probability that a flammable concentration is present. In IEC terminology, these

three levels are Zones 0, 1, and 2. North American Division 1 includes both Zone 0 and Zone 1, and North American Division 2 is basically equivalent to Zone 2. Though the definitions of zones are similar in almost all standards, the application of the words to specific industrial situations is different.

4.2.4 Temperature classification

Prior to 1971 the autogenous ignition (or autoignition) temperature, AIT, was a criterion for group classification. Inclusion of the AIT as one of the classification criteria caused problems for those trying to classify new materials that had not been tested, because other flammability and combustion parameters of flammable gases and vapors are not correlated to AIT. For example, the AIT of diethyl ether is 160°C (320°F). Hydrogen has an AIT of approximately 520 °C (968 °F). Methane has an AIT of approximately 630 °C (1166 °F). Yet hydrogen is much more easily ignited by an arc than diethyl ether. Methane is much less easily ignited. Hydrogen requires very close-fitting flanges to prevent transmission of an explosion, but the flanges for an enclosure to protect against transmission of an explosion in diethyl ether may be much more widely separated, i.e., the MESG of diethyl ether is several times that of hydrogen.

When the 1971 *National Electrical Code* and the *Canadian Electrical Code* removed AIT as one of the criteria for material classification, the practice of temperature marking was introduced. Table 1 lists the temperature codes recognized in the NEC. Enclosures containing heat-producing devices must be marked by a temperature code or with the maximum surface temperature of the enclosure based on 40°C (104°F) ambient. Those that do not have an alphabetical suffix, i.e., T1-T6, are recognized internationally by the International Electrotechnical Commission (IEC), by CENELEC, and by many national standards bodies. In the United States and Canada, equipment of the nonheat-producing type (such as junction boxes, conduit, and fittings) and equipment of the heat-producing type (such as industrial process transmitters and transducers) having a maximum temperature not more than 100 °C (212 °F) need not be marked. The temperature classification marking also applies to surfaces other than those of the enclosures in the case of intrinsically safe and nonincendive equipment.

Table 1 — Temperature identification numbers

Maximum Surface Temperature		
Degrees C	**Degrees F**	**Identification Number**
450	842	T1
300	572	T2
280	536	T2A
260	500	T2B
230	446	T2C
215	419	T2D
200	392	T3
180	356	T3A
165	329	T3B
160	320	T3C
135	275	T4
120	248	T4A
100	212	T5
85	185	T6

4.2.5 Grouping of materials

4.2.5.1 A hazard grouping of materials is always relative to a stated property, i.e., to a particular ignition mechanism or a means of hazard reduction. Materials that are very much alike relative to ignition by electrical arcs or materials that have similar MESG may behave quite differently with respect to ignition by a hot surface.

4.2.5.2 Table 2 compares several countries'/organizations' designations of gas groups. At the present time most national standards use the IEC group designations, where "II" indicates an above-ground facility and "I" indicates a hazard due to methane in the below-ground works of a mine. The comparisons of Table 2 are approximate. For example, North American Group C is approximately the same list of materials as IEC Group IIB. Grouping is an arbitrary designation of dividing lines in a continuous series of values of a particular parameter.

Table 2 — Comparison of classification of flammable vapors and gases (approx.).

IEC (60079-12)	UK (BS4683)*	UK (BS229: 1957)*	Germany (VDE0171) *		USA National Electrical Code Group		Typical gases and vapors
IIA	IIA	II	1		D		Ethane, propane, butane, pentane, hexane, methane, heptane, octane, nonane, decane, acetic acid, acetone, methanol, toluene, ethyl acetate
IIB	IIB	III	2		C	**Class I**	Ethylene, coke oven gas, dimethyl ether, diethylene, ethylene oxide
IIC	IIC	IV	3n	3a	B		Hydrogen
				3b			Carbon disulphide
				3c	A		Acetylene
							Ethyl nitrate

* Prior to CENELEC

5 Protection techniques for electrical apparatus in hazardous (classified) locations

The most basic protection technique is to avoid placing electrical equipment in a hazardous (classified) location. Facility planning should take this factor into account, leaving only those situations where there is no alternative.

Three basic methods of protection are:

a) explosion confinement;

b) isolation of the ignition source; and

c) energy release limitation.

Within each basic method, one (or more) specific technique necessitates specialized design in order to minimize the potential risk of operating electrical apparatus in hazardous (classified) locations.

5.1 Explosion confinement and flame quenching

5.1.1 Explosionproof (similar to the international term *flameproof*)

Explosionproof, applicable to Class I areas only, is a specific protection technique in which the ignition source, fuel, and oxygen may coexist and ignition may occur. However, any ignition that does occur is confined within an enclosure strong enough to withstand any explosion pressure associated with the gas groups for which it is designated. Also, all joints have close enough values and tolerances so that flame, sparks, or escaping hot gases are cooled sufficiently to prevent the external atmosphere from being ignited. Additionally, all external surfaces must be kept below the autoignition temperature for the specific

gases or vapors involved. The explosionproof technique is gas-group dependent – i.e., an enclosure designed and tested for Class I Group C would not be suitable for use in a Group B atmosphere.

5.1.2 Powder filling

Powder filling is a technique whereby the electrical components are immersed in a powder to a depth sufficient to ensure that any arcing below the powder cannot ignite the flammable atmosphere surrounding the apparatus.

The concept was developed in France and generally referred to sand, quartz, or solid glass particles used as filling materials. At the present time, however, only quartz is permitted. Refer to ISA S12.25.01 (IEC 60079-5 Mod).

5.2 Isolation from flammable atmospheres

Isolation of the ignition source from the flammable atmosphere may be accomplished by several techniques. Some of the most common techniques include pressurization, purging, continuous dilution, and inerting.

5.2.1 Pressurization

Pressurization reduces the concentration of any flammable gas or vapor initially present to an acceptably safe level and isolates electrical components from the external atmosphere by maintaining a pressure within the equipment enclosure higher than that of the external atmosphere. Thus, the external atmosphere is prevented from entering the enclosure. Unlike explosionproofing, the purging technique is not gas-group dependent with the following exceptions:

a) For Type Y, a nonincendive component and nonincendive circuits must be gas-group related.

b) For Type X, door interlock and purge fail power cutoff must be rated for the location.

In the case of enclosures in which flammable materials are intentionally introduced within the enclosure (such as with gas analyzers), a different technique, commonly referred to as continuous dilution, is required. Refer to NFPA 496, although this standard does not use the term CONTINUOUS DILUTION. Also refer to 5.2.2.

5.2.1.1 In North America the pressurization technique is used for reducing the classification within the enclosure to a lower level, such as from Division 1 to Division 2 or nonhazardous (unclassified) or from Division 2 to nonhazardous (unclassified). The European and North American requirements agree in principle, but the bases for the respective requirements are equipment construction criteria. The end result is essentially the same. Reference IEC 60079-2 and 60079-13.

A discussion of three different sets of requirements dependent upon the area classification and the nature of the enclosed apparatus follows.

United States and Canada

Type X Pressurization. In the United States and Canada, Type X pressurization is a method of reducing the classification within an enclosure from Division 1 to nonhazardous (unclassified). Type X purging requires that the enclosure pressure be monitored and that the electrical power be mechanically disconnected upon loss of positive pressure.

Type Y Pressurization. In the United States and Canada, Type Y pressurization is a method of reducing the classification within an enclosure from Division 1 to Division 2. Devices that employ Type Y

pressurization must be suitable for use in Division 2 locations without pressurization. A visual or audible warning is required for loss of positive pressure.

Type Z Pressurization. In the United States and Canada, Type Z pressurization is a method of reducing the classification within an enclosure from Division 2 to nonhazardous (unclassified). A visual or audible warning is required for loss of positive pressure.

International

Types X, Y, and Z pressurization are not currently recognized outside the United States and Canada, although the concept of Type X, Y, and Z is proposed in the 1997 draft revisions to IEC 60079-2. The IEC currently uses the designation “first case” for Type X and Type Y pressurization and “second case” for Type Z pressurization. International requirements for visual and/or audible alarms for loss of enclosure pressure, when the pressurization technique is used, vary from country to country. Loss of pressure may also require power disconnection.

5.2.1.2 Pressurization for Class II

Pressurization for Class II is the technique of supplying an enclosure with clean air or an inert gas, with or without continuous flow, at sufficient pressure to prevent the entrance of combustible dusts.

5.2.2 Continuous dilution

Continuous dilution is a derivation of the purging technique and is intended for electrical equipment enclosures in which a flammable material is deliberately introduced. Such equipment may include gas analyzers, chromatographs, and similar instruments. The principle involved is to introduce sufficient flow of protective gas to dilute any flammable gas present during normal operating conditions or failure conditions to a level well below the lower explosive limit (normally, 25 percent of LFL/LEL). An example of a failure condition would be a broken tube transporting flammable gas. As with purging, there are three types of protection, depending upon the conditions of release within the enclosure. The safeguards include (1) monitoring the presence of the protective gas, (2) removing electrical power, and (3) alarming — depending on the conditions of internal release and the nature of the enclosed electrical components. Safeguards depend upon whether or not the electrical parts are normally a source of ignition or meet the requirements for operation in a Division 2 hazardous (classified) location. A continuous dilution system may also be used as a purging or pressurization system to prevent any external flammable gas or vapor or combustible dusts from entering the enclosure.

5.2.3 Oil Immersion

While oil immersion is not a common protection technique for instruments, it is an acceptable isolation method. The most common application is for electrical equipment such as switches, relays, and transformers. All electrical parts are submersed in either nonflammable or low-flammability oil, which prevents the external atmosphere from contacting the electrical components. The oil often serves also as a coolant. Refer to ISA S12.26.01 (IEC 60079-6 Mod).

5.2.4 Sealing (sealed device)

Sealing is a technique primarily applicable to Division 2 classified areas. The basic principle provides for the isolation of electrical components within an enclosure by sealing the enclosure well enough to prevent the casual entrance of any external flammable atmosphere. Sealing may be accomplished by several means, from simple gasketing to a glass-to-metal hermetic seal. Reference ISA-S12.12, IEC 60079-15, and CEC 22.2, No. 213, for further information.

A hermetic seal is considered effective enough to be insensitive to gas group, but a gasketed enclosure is sensitive to particular gases/vapors (based on their diffusion constants and on the effectiveness of the gasket seal). For example, hydrogen, with its small molecular structure, will diffuse much more easily than the heavier hydrocarbons.

The sealing technique has been applied in Europe to a variety of process control equipment. In North America, hermetically sealed components such as relays, push button contacts, and limit, level, and pressure switches are commonly used.

The following are two levels of protection provided by sealing: (1) the enclosure is sealed, and (2) application is restricted to areas that become hazardous only upon equipment or process failure (Division 2).

5.2.5 Encapsulation

Potting or casting are both isolating techniques in which the electrical parts are encapsulated in a solidified electrically insulating material, preventing the flammable atmosphere from contacting the electrical components. Most encapsulation has been for the purpose of isolating hot components from the atmosphere in order to obtain a lower temperature rating or to permit reduced creepage distances because the spacings are shielded from conductive contamination. Reference ISA-S12.23.01 (IEC 60079-18 Mod).

5.2.6 Inert gas filling

Inert gas filling is a technique of filling the interior of an enclosure with an inert gas. It typically is used in conjunction with sealed or pressurized enclosures. Refer to 5.2.4.

5.3 Energy release limitation

5.3.1 Intrinsic safety

The application of intrinsic safety is limited to equipment and circuits in which the available energy required for operation is inherently low. Intrinsic safety involves the limitation of the available energy in a circuit to a level at which any spark or thermal effect is incapable of causing ignition of a flammable atmosphere under test conditions that include the application of circuit and component faults. As a result, the technique is widely used in the instrument industry, e.g., 4 to 20 mA signal circuits; temperature, flow, pressure, and level measurement instruments; portable battery-operated instruments (radios, combustible gas detectors). Certain fault conditions need to be considered in the design and evaluation. For installation information, refer to ISA-RP12.6 and NEC Article 504.

5.3.2 Nonincendive equipment (internationally referred to as energy limited apparatus and circuits)

The nonincendive approach is similar to the intrinsic safety approach in basic principle but differs greatly in detail. There are two major differences. First, nonincendive circuits are evaluated under normal conditions only (i.e., no fault conditions need to be considered). Equipment meeting the nonincendive criteria is suitable for use only in Division 2 areas in which the atmosphere is normally nonflammable and requires a breakdown in the process or the process equipment to make it flammable.

Second, relative to the components used, few detailed requirements must be met other than those applicable to nonhazardous (unclassified) location use as related to personnel shock and fire hazard. A typical analysis involves itemizing all parts that could potentially interrupt a circuit such as switches, relays, connectors, and potentiometers. These components are then analyzed or tested to see if they can ignite the specified flammable atmosphere.

5.4 Other methods of protection

5.4.1 Restricted breathing

Restricted breathing is a technique developed by the Swiss. It can be considered to be a modified form of sealing. In the Swiss utilization of the technique, fairly large enclosures of relays and other ignition-capable apparatus are gasketed so they are tightly sealed. The protection principle employed is that the enclosure is sufficiently tight to make it highly unlikely that a flammable cloud of gas would surround the enclosure for the length of time necessary for enough flammable material to enter the enclosure that a flammable concentration would exist in the enclosure. This technique is applicable only to Zone 2 locations. Refer to IEC 60079-15 and ISA S12.12.01.

5.4.2 Increased safety

Increased safety is a technique developed in Germany. It has been used for equipment such as terminals, motors, and lighting fixtures. The motors are specially designed to remain below the autoignition temperature, even under certain locked rotor conditions, and have special connection facilities designed to prevent loosening even under severe vibration. Refer to IEC 60079-7 and ISA S12.16.01.

5.4.3 Dust-ignitionproof

Dust-ignitionproof enclosures are essentially sealed enclosures that prevent the entrance of dusts. Additionally, the outer enclosure temperature must be maintained below specific limits. For additional information, refer to UL 1203.

5.5 Summary of Types of Protection

Table 3 provides a summary of various Types of Protection and locations in which they are permissible.

Table 3 — Summary of Types of Protection (flammable gases or vapors-in-air mixtures)

Designation	Technique	Zone *
d	Flameproof enclosure	1
e	Increased safety	1
ia	Intrinsic safety	0
ib	Intrinsic safety	1
[ia]	Intrinsically safe associated apparatus	Non-hazardous
[ib]	Intrinsically safe associated apparatus	Non-hazardous
m	Encapsulation	1
nA	Non-sparking equipment	2
nC	Sparking equipment in which the contacts are suitably protected other than by restricted breathing enclosure.	2
nR	Restricted breathing enclosure	2
o	Oil immersion	1
p	Purged and pressurized	1 or 2
q	Powder-filled	1

* Does not address use where a combination of techniques is used.

For protection techniques (Types of Protection) applicable to Classes I, II, and III; Division 1 and 2, refer to Article 500-4(a) of the NEC.

5.5.1 A special protection category, *Type s*, is a technique other than those that have been standardized. When an area is classified Zone 0, it is common in some countries to provide two or more protection techniques, such as pressurizing a flameproof enclosure. *Ex s* has also been applied for Zone 1, where, for example, a transmitting device partly satisfies the increased safety requirements and also partly satisfies the intrinsic safety requirements. The combination results in a device that is safe for use in a hazardous (classified) location but does not satisfy a specific set of requirements for a single protection technique.

5.5.2 The intrinsic safety column indicates that there are two sets of requirements — *ia* and *ib*; *ia* is intended for Zone 0 applications while *ib* is for Zone 1 applications. The difference in requirements is that *ia* considers two simultaneous faults while *ib* considers only one. The approach of the United States and Canada (in the Division system) uses the two-fault criteria for all intrinsic safety applications since a Division 1 area classification includes both the Zone 0 and Zone 1, and equipment must be rated to the most stringent (Zone 0) requirements.

6 Wiring methods

For the installation of electrical apparatus, three basic installation systems are allowed:

a) Conduit systems (Reference Figures 11, 13, and 15 and Tables 4a and 4b)

b) Cable systems with indirect entry (Reference Figures 5, 6, 12, 14, and 15 and Tables 4a and 4b)

c) Cable systems with direct entry (Reference Figure 7, 8, 9, 12, 14 and 15 and Tables 4a and 4b)

NOTE — INTRINSICALLY SAFE ELECTRICAL APPARATUS and NONINCENDIVE FIELD WIRING can be installed with less restrictive wiring methods. See NEC Section 504-20 or 501-4(b) Exception, respectively.

In the United States, NEC Article 501-4 allows only 1) a conduit system, 2) mineral-insulated (Type MI) cables, or 3) in industrial establishments with restricted public access where the conditions of maintenance and supervision ensure that only qualified persons will service the installation, Type MC cable, listed for use in Class I, Division 1 locations, with a gas/vaportight continuous corrugated aluminum sheath, an overall jacket of suitable polymeric material, separate grounding conductors in accordance with Section 250 and provided with termination fittings listed for the application.

NEC Article 501-4(b) allows for the use of certain other types of cables in Division 2 areas.

The European harmonized standards, EN 50014 through EN 50039 permit apparatus to be designed that can be installed using one or more of the three installation approaches. The installation requirements, however, are different from country to country.

NOTE — All figures referred to in this section are shown below.

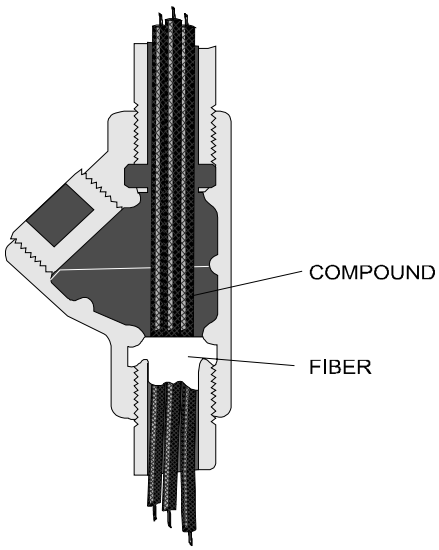


Figure 1 — Vertical conduit seal

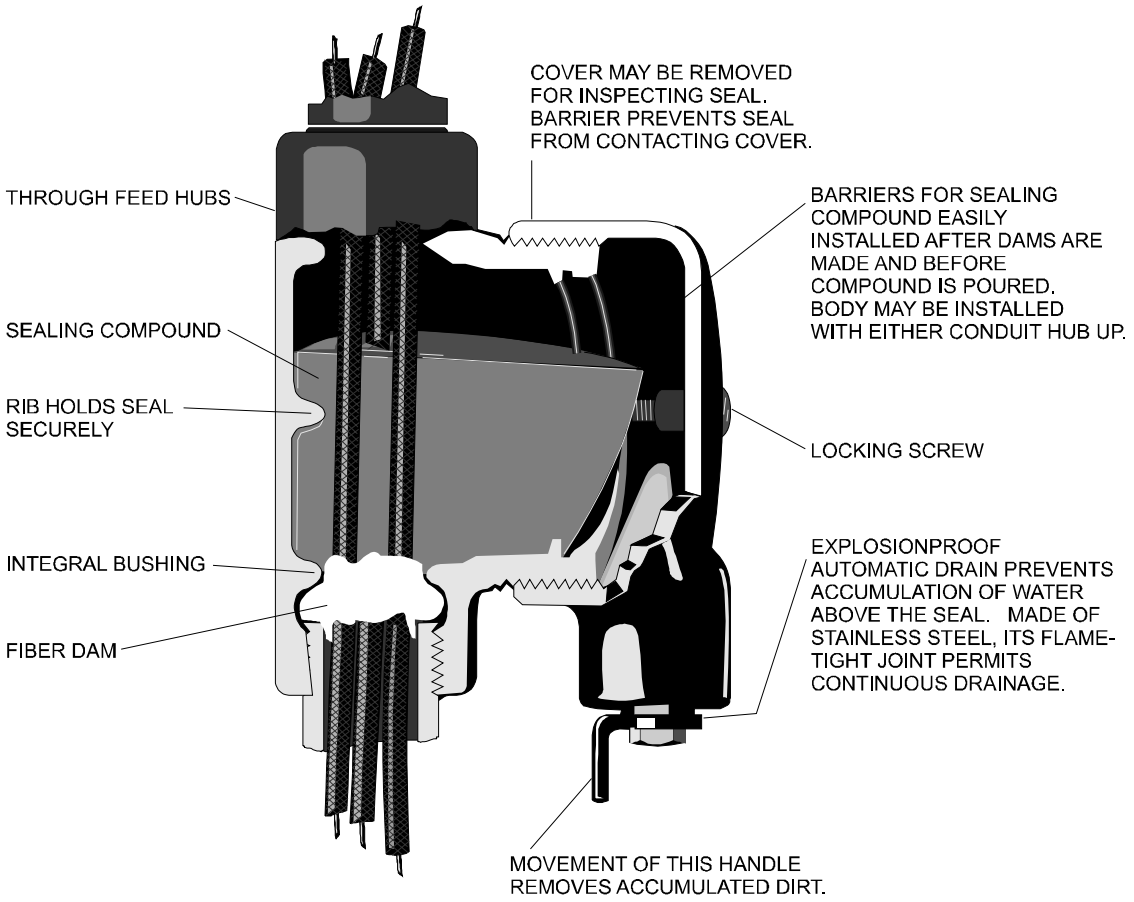


Figure 2 — Conduit drain seal

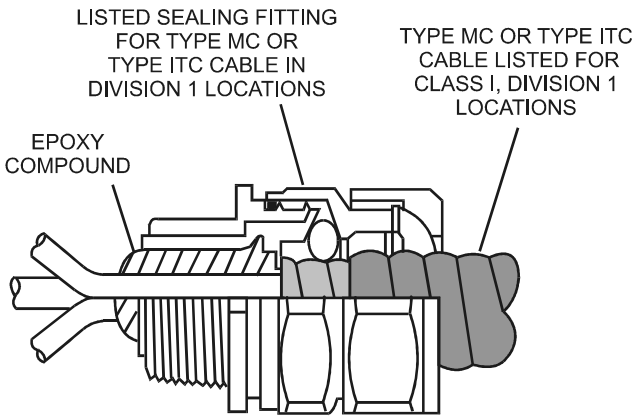


Figure 3 — Cable seal

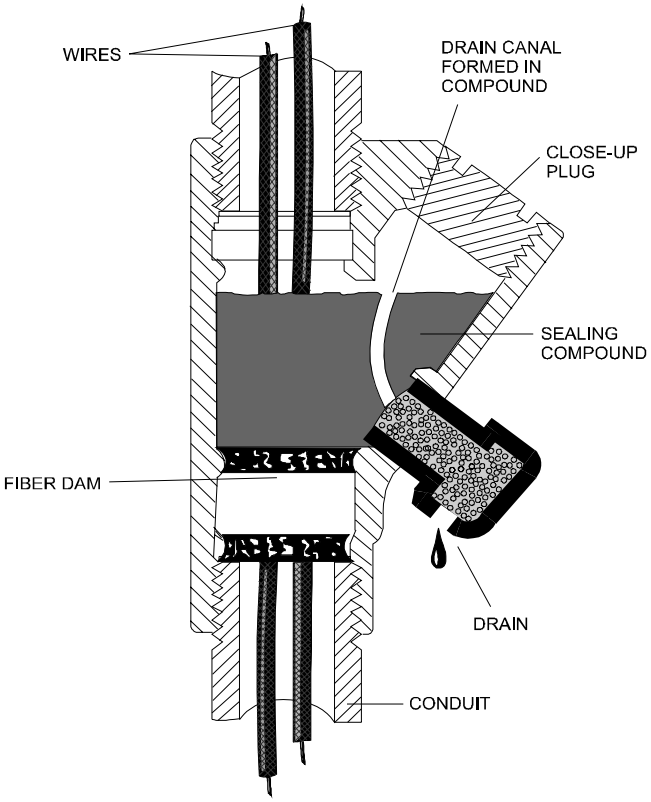


Figure 4 — Conduit drain seal

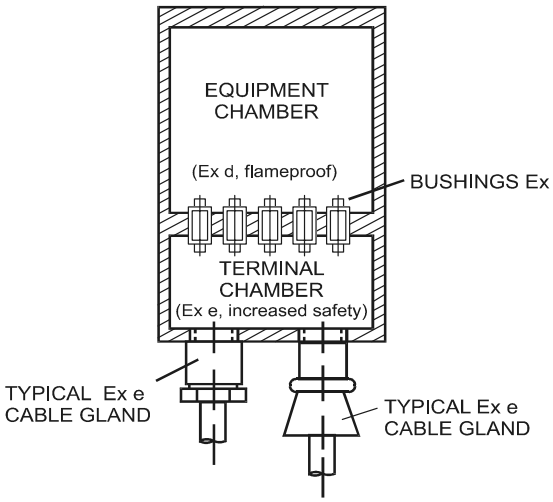


Figure 5 — Cable system (indirect entry)

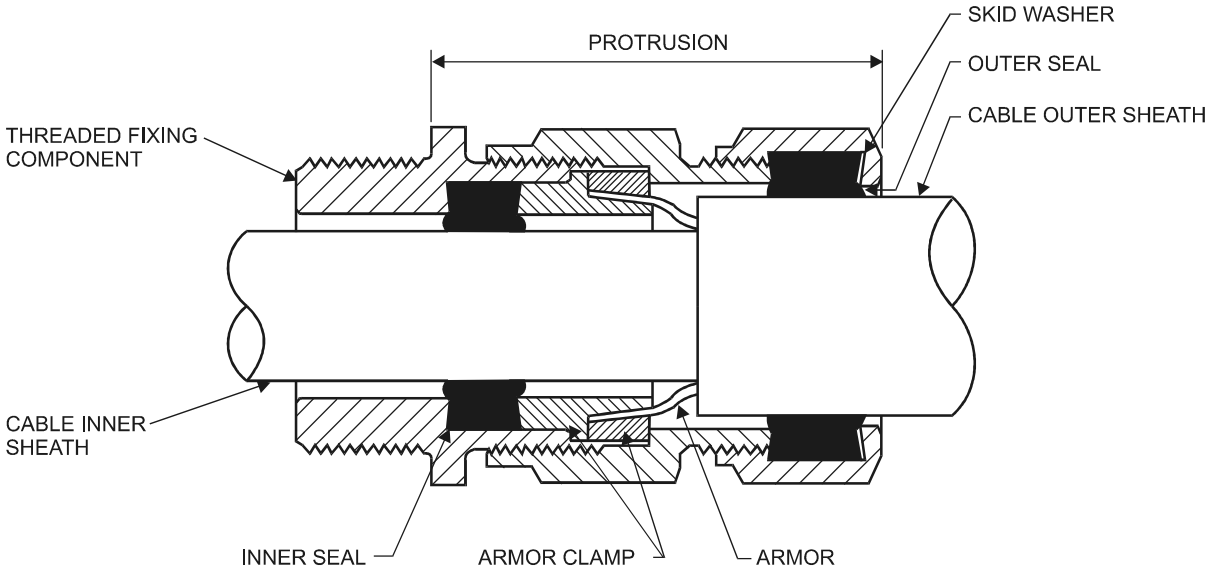


Figure 6 — Cable gland (indirect entry)

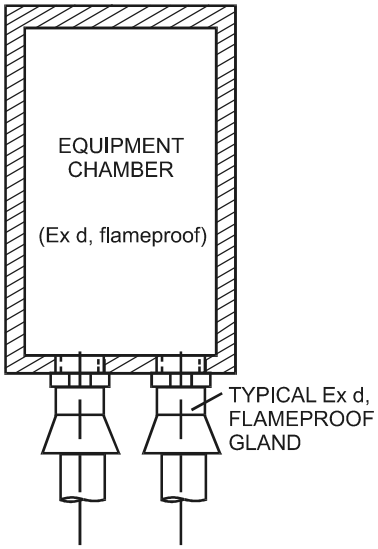


Figure 7 — Cable system (direct entry)

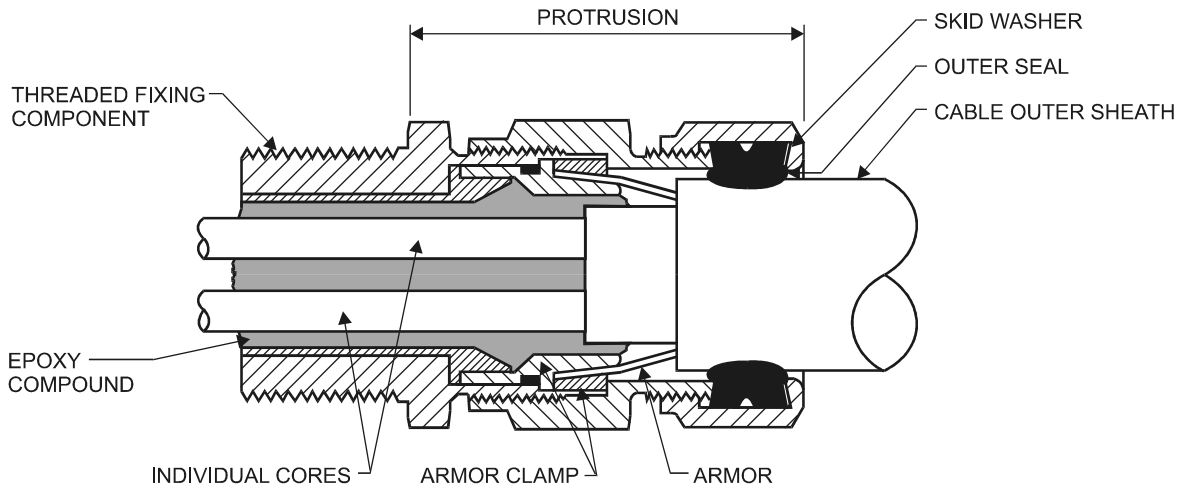


Figure 8 — Cable gland (direct entry)

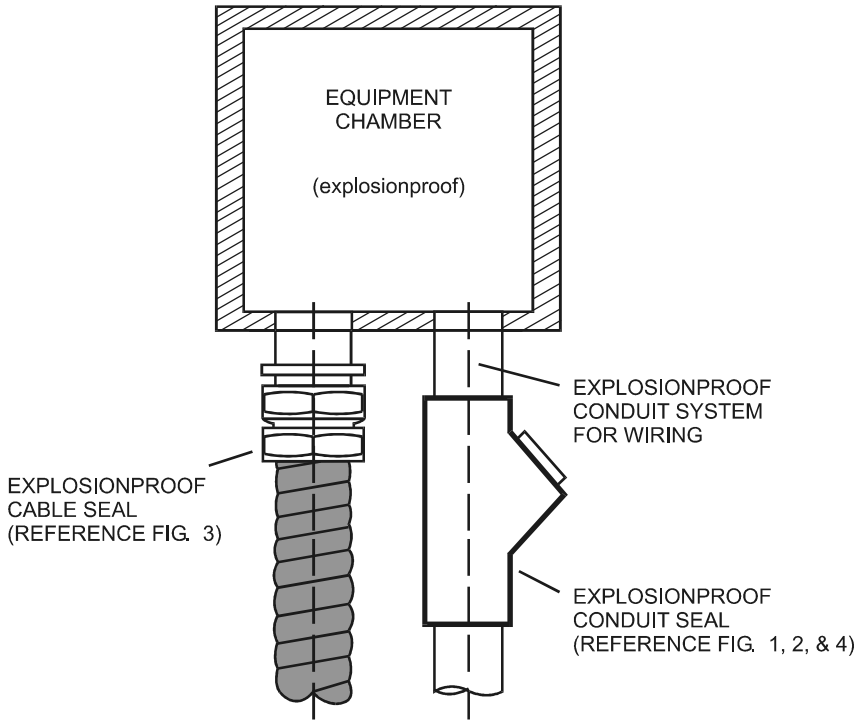


Figure 9 — Conduit system (direct entry)

6.1 Conduit system

6.1.1 General

With conduit systems in Class I, Division 1, locations in the United States, the electrical wiring is installed inside closed threaded metal pipes (rigid steel or intermediate metal conduit) approved for the purpose. The pipes, in turn, are screwed into entrances in the enclosures, which contain electrical equipment. The entire conduit system, including all fittings, is required to be explosionproof and frequently requires an explosionproof seal between the connected enclosure and the pipe. In Class I, Division 2, locations, the conduit system need be explosionproof only between any explosionproof enclosure and the required sealing fittings.

In Class I, Division l, locations in Canada, one difference is that threaded intermediate metal conduit is not acceptable.

In this section all references to *seal* and *sealing* refer to an approved conduit or cable seal that is filled with suitable compound, is designed to contain an explosion in the enclosure to which it is attached, and is approved for use in Class I locations [Figures 1, 2, 3, and 4] (see Article 501 of the NEC).

6.2 Cable systems

In the United States, in Class I, Division 1, locations, the NEC will allow 1) mineral-insulated Type MI cable, 2) Type MC or Type ITC cable listed for use in Class I, Division 1, locations, with a gas/vaportight continuous corrugated aluminum sheath, an overall jacket of suitable polymeric material, separate grounding conductor in accordance with Section 250, and provided with termination fittings listed for the application (Figure 3), and 3) cables used in intrinsically safe systems as permitted by Article 504 of the NEC. For Class I, Division 2 locations, certain cable constructions are allowed. Refer to Table 4a.

Internationally, openly installed cable systems are common, using heavy-duty sheathed cables (i.e., with an outer sheath of rubber, plastic, or metal). Armored or braided cable is often required in Zone 1 or in areas where damage might occur to unprotected cable. The metal braid or armor is covered by an outer sheath of rubber, plastic, or other synthetics, and the braid or armor is grounded.

Different technologies have been developed in various countries regarding the entry of cables into explosionproof (flameproof) enclosures. In the course of harmonizing the standards, all these entry possibilities were included in the CENELEC standards.

6.2.1 Indirect Entry - In Germany, and in many other countries influenced by German technology, only indirect entry (via a terminal chamber that provided increased safety protection) and factory potted and installed flameproof cable entries were allowed.

Cables enter the terminal chambers via cable glands and connect to increased safety terminals (Figures 5 & 6). Then the single conductors enter into the flameproof chamber via post-type bushings or conductor bushings. The installer needs only to open the terminal chamber of increased safety for connection, not the flameproof chamber. Flameproof cable glands are not required.

6.2.2 Direct Entry - Cables enter the flameproof chamber directly. For cable entry, suitable flameproof cable glands must be used. (Figures 7 & 8)

6.3 Conduit and cable seals

Seals are installed in conduit and cable systems (except for Type MI cable systems with approved explosionproof terminations) to

a) confine an explosion occurring in an enclosure or a conduit system to only that enclosure or that portion of the conduit system;

b) minimize the passage of gases, vapors, or liquids and prevent the passage of flames through the conduit or cable system from a classified to an unclassified area, or from one enclosure to another;

c) prevent *pressure piling* — the buildup of pressure inside conduit systems (ahead of an explosion's flame front) caused by precompression as the explosion travels through the system. Exploding precompressed gases may reach pressures that would exceed the design pressure of the enclosures.

6.3.1 Seal requirements

6.3.1.1 Enclosure entries

In Division 1 and Division 2 locations, a seal must be installed in every conduit or cable system (reference NEC Article 501-5 for requirements) that enters an enclosure containing arcing, sparking or high temperature apparatus in which the enclosure is required to be explosionproof.

Multiconductor cables should be sealed in an approved fitting only after removing the jacket and any other coverings so the sealing compound will surround each individual insulated conductor and the outer jacket of the cable. The multiconductor cable may be treated as a single conductor if the cable end is sealed by an approved means within the enclosure.

The conduit system between an enclosure and the required seal must be explosionproof, even in Division 2 locations, since the conduit system must be able to withstand the same internal explosion as the enclosure to which it is attached. In Division 1 and Division 2 locations, approved explosionproof unions, couplings, elbows, reducers, and conduit bodies similar to "L," "T," or "Cross" types are the only fittings allowed between the sealing fitting and the enclosure. The conduit body cannot be larger than the trade size of its associated conduit.

In addition to the above, in Division 1 areas only, explosionproof seals must be installed in each two-inch or larger conduit run entering an enclosure that contains splices, taps, or terminals. All seals must be installed within 18 inches (457 mm) from the enclosures to which they are attached.

6.3.1.2 Process Instruments

Seals and drains shall be installed in each conduit or cable connection to devices installed on process lines that contain flammable fluids and depend upon a single compression seal, diaphragm, or tube (such as a Bourdon tube) as a barrier between the process fluid and the conduit or cable. This is to prevent flammable process fluids from entering conduit or cable systems and being transmitted to unclassified areas or to electrical arcing or high temperature devices in other portions of the system if the process seal fails. The additional seal and the interconnecting enclosure or conduit or cable system shall meet the temperature and pressure conditions to which they will be subjected upon failure of the compression seal. Ordinary conduit or cable seals typically may not meet this criterion due to their leakage rate. Typical examples of such devices are solenoid valves and pressure, temperature, and flow switches or transmitters. This requirement applies even in unclassified areas. Draining provisions must be such that process-line leaks past the process seal will be obvious.

6.3.1.3 Classified area boundaries

Wherever a conduit run passes from a Division 1 to a Division 2 area, from a Division 2 to an unclassified area, or any other combination thereof, a seal must be placed in the conduit run at the boundary, on either side. The conduit system must not contain any union, coupling, box, or other fitting between the sealing fitting and the point at which the conduit leaves the Division 1 or Division 2 area. An exception to the

above is that an unbroken rigid metal conduit that passes completely through a Division 1 or a Division 2 area is not required to be sealed if the termination points of the unbroken conduit are in unclassified locations and the conduit has no fitting less than 12 inches (305 mm) beyond each boundary.

If drain seals are used at an area classification boundary, care must be exercised in the placement of such seals to ensure that gases or vapor cannot be communicated across the boundary through the conduit system by way of the seal's drain passage. Reference Figure 10.

Cables with an impervious continuous sheath do not have to follow the same sealing requirements as conduit systems when crossing Division 2 unclassified area classification boundaries. Such cables are not required to be sealed unless the cable is attached to process equipment or devices that may cause a pressure in excess of 6 inches of water (1.5 kPa) to be exerted at a cable end. In this case, a seal or other means shall be provided to minimize migration of flammables into an unclassified area or to arcing or high temperature devices in other portions of the system, in accordance with NEC Article 501. No seal is then required at the boundary location. Cables with an unbroken, impervious, continuous sheath are permitted to pass through a Division 2 unclassified area classification boundary without seals.

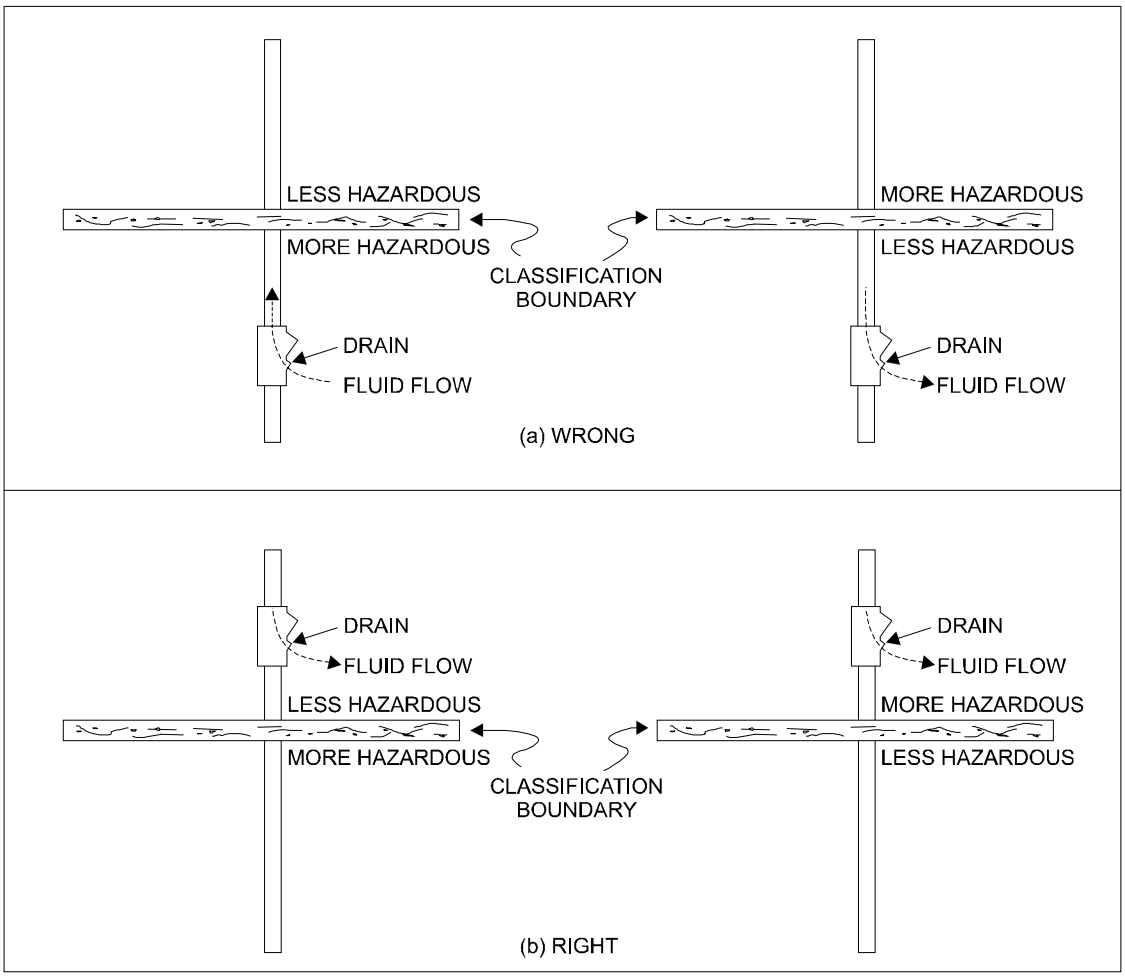


Figure 10 — Placement of drain seals (Reference API RP 14F, Figure 7, Recommended Practice for Design and Installation of Electrical Systems for Fixed and Floating Offshore Petroleum Facilities for Unclassified and Class I, Division 1 and Division 2 Locations)

6.3.1.4 Installation

In addition to being placed in proper locations, the following practices should be observed when installing sealing fittings:

a) Sealing fittings must be accessible.

b) Sealing fittings must be mounted only in the positions for which they were designed. Some seals are designed only to be installed vertically; some can be installed either vertically or horizontally; a third type can be installed in any position.

c) Pouring hubs must be properly oriented. The hub through which the sealing compound is to be poured must be installed above the sealing cavity to properly pour the seal.

d) Only a sealing compound and fiber approved for a particular sealing fitting should be used, and the manufacturer's instructions should be followed for the preparation of dams (if applicable) and the preparation and installation of the sealing compound. Particular attention should be paid to temperature limitations of the sealing compound during installation.

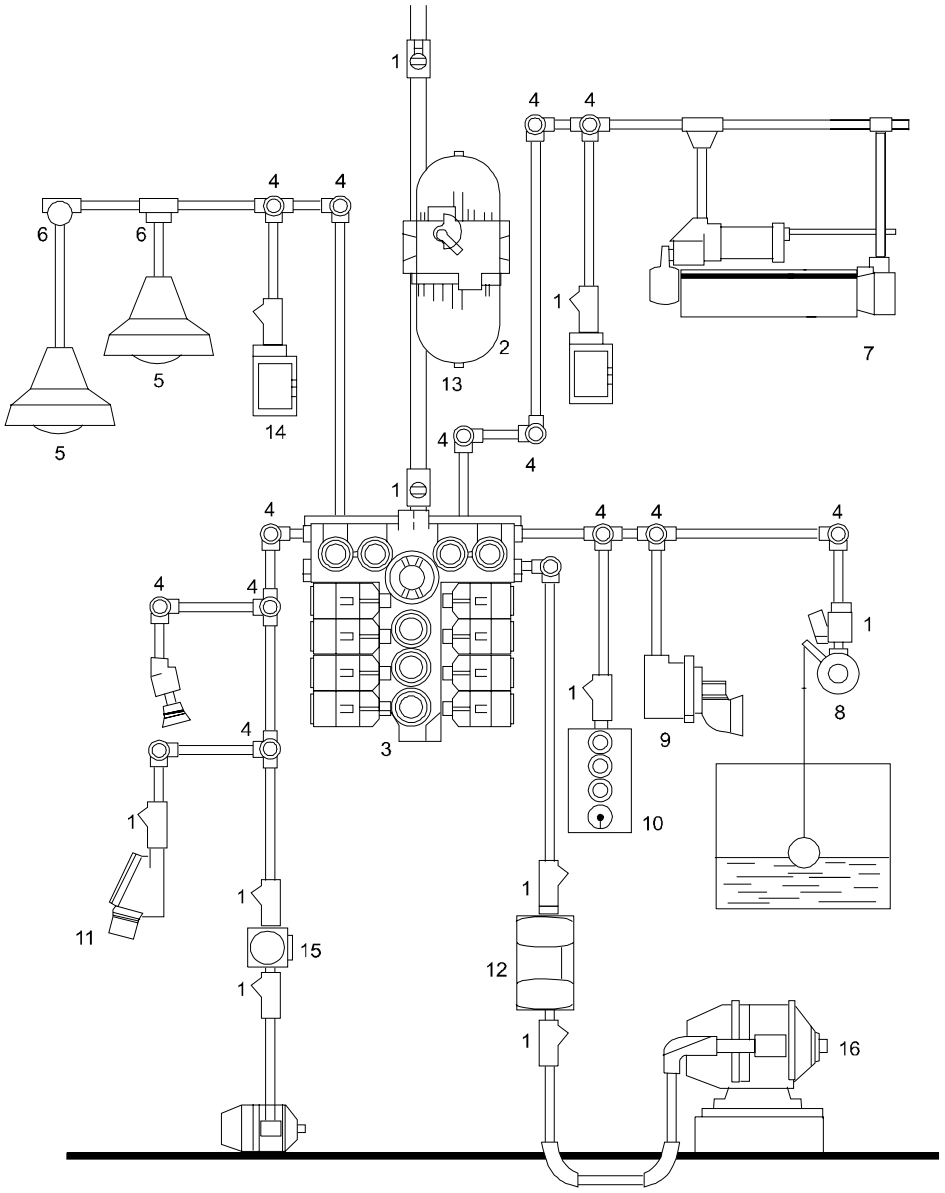
e) No splices or taps are allowed in sealing fittings. Sealing compounds may not be insulation materials and may absorb moisture, causing grounding of the circuit conductors.

f) Sealing fittings with drain provisions should be installed at the low points of a conduit system to allow drainage of conduit where water or fluids may accumulate in the conduit/system.

g) Factory-sealed devices such as toggle switches, push buttons, lighting panels, and lighting fixtures eliminate the need for externally sealing those particular devices.

6.4 Comparison of the installation systems

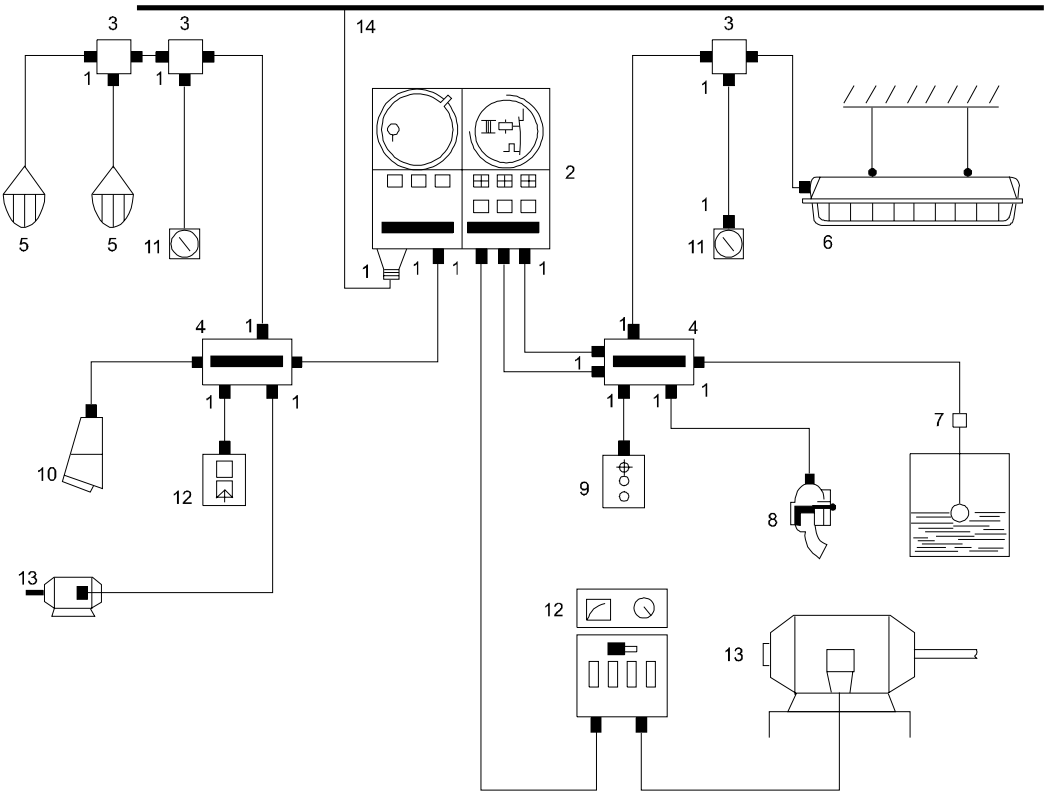
Figure 11 depicts a typical international Group II, Zone 1 conduit system installation.



1 SEAL
2 ISOLATOR
3 CONTROL/DISTRIBUTION BOARD WITH FACTORY SEALED SUBCIRCUITS
4 EXPLOSIONPROOF JUNCTION BOX
5 PENDANT LIGHT FITTING
6 EXPLOSIONPROOF JUNCTION BOX, SERVING AS SUPPORT FOR PENDANT LIGHT FITTINGS
7 FLUORESCENT LIGHT FITTING
8 FLOAT SWITCH
9 ALARM
10 CONTROL AND INDICATING EQUIPMENT
11 SOCKET
12 BREATHER
13 DRAIN
14 INSTALLATION SWITCH
15 MOTOR PROTECTION CIRCUIT BREAKER
16 EXPLOSIONPROOF MOTOR

Figure 11 — Typical international Group II, Zone 1 conduit system installation

Figure 12 depicts a typical international Group II, Zone 1 cable system for a similar installation.



1 CABLE GLANDS
2 FLAMEPROOF CONTROL PANEL Ex d WITH TERMINAL CHAMBER Ex e, BUILT-IN MAIN INDICATOR, CONTROL PUSH BUTTON INDICATING LIGHT
3 JUNCTION BOXES OF "INCREASED SAFETY" TYPE Ex e
4 DISTRIBUTION TERMINAL BOXES Ex e
5 PENDANT LIGHT FITTINGS Ex e OR Ex d
6 FLUORESCENT LIGHT FITTINGS Ex e OR Ex d
7 FLOAT SWITCH "FLAMEPROOF ENCLOSURE" OR IN STANDARD DESIGN CONNECTED TO AN INTRINSICALLY SAFE CIRCUIT
8 ALARM
9 CONTROL BOX Ex e WITH MODULES Ex d
10 SOCKET Ex d
11 LIGHT SWITCH Ex d
12 MOTOR CONNECTION CONTROL BOX Ex e WITH MODULES Ex d
13 MOTORS Ex e OR Ex d
14 CABLES

Figure 12 — Typical international Group II, Zone 1 cable system installation

Figure 13 depicts a typical United States and Canadian Class I, Division 1 conduit system installation.

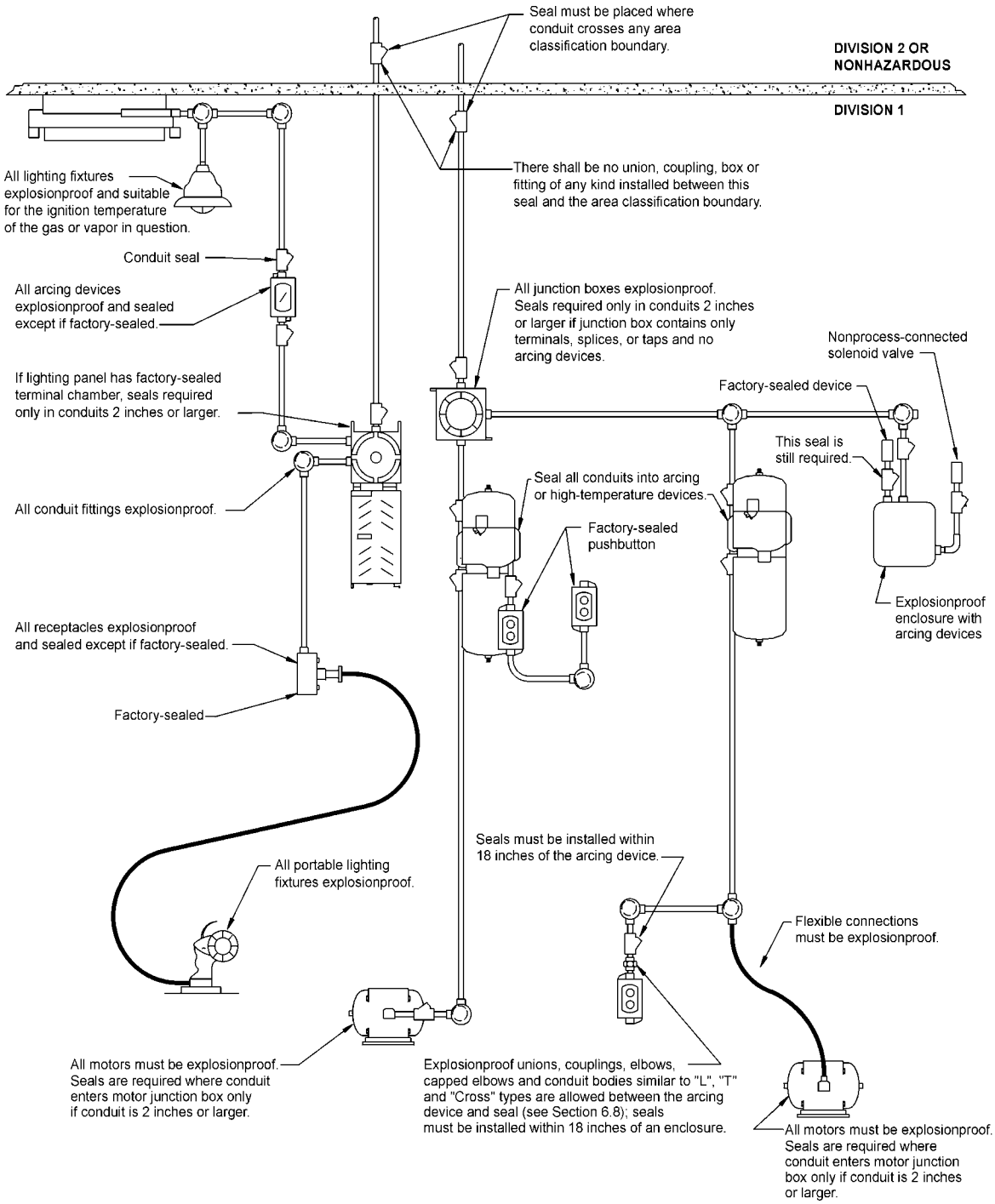
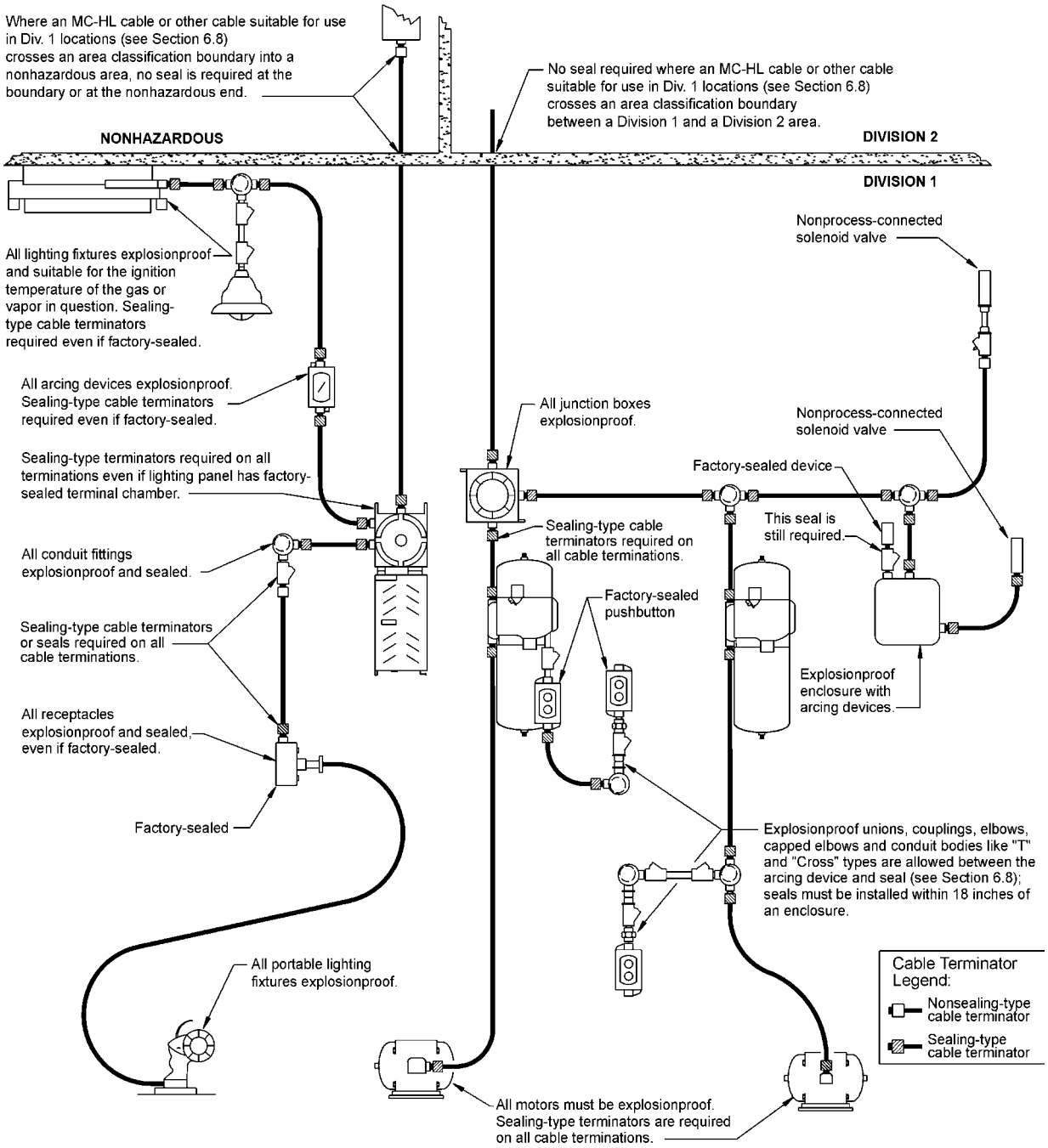


Figure 13 — Typical United States and Canadian Class I, Division 1 conduit system installation (Reference API RP 14F, Figure 1, Recommended Practice for Design and Installation of Electrical Systems for Fixed and Floating Offshore Petroleum Facilities for Unclassified and Class I, Division 1 and Division 2 Locations)

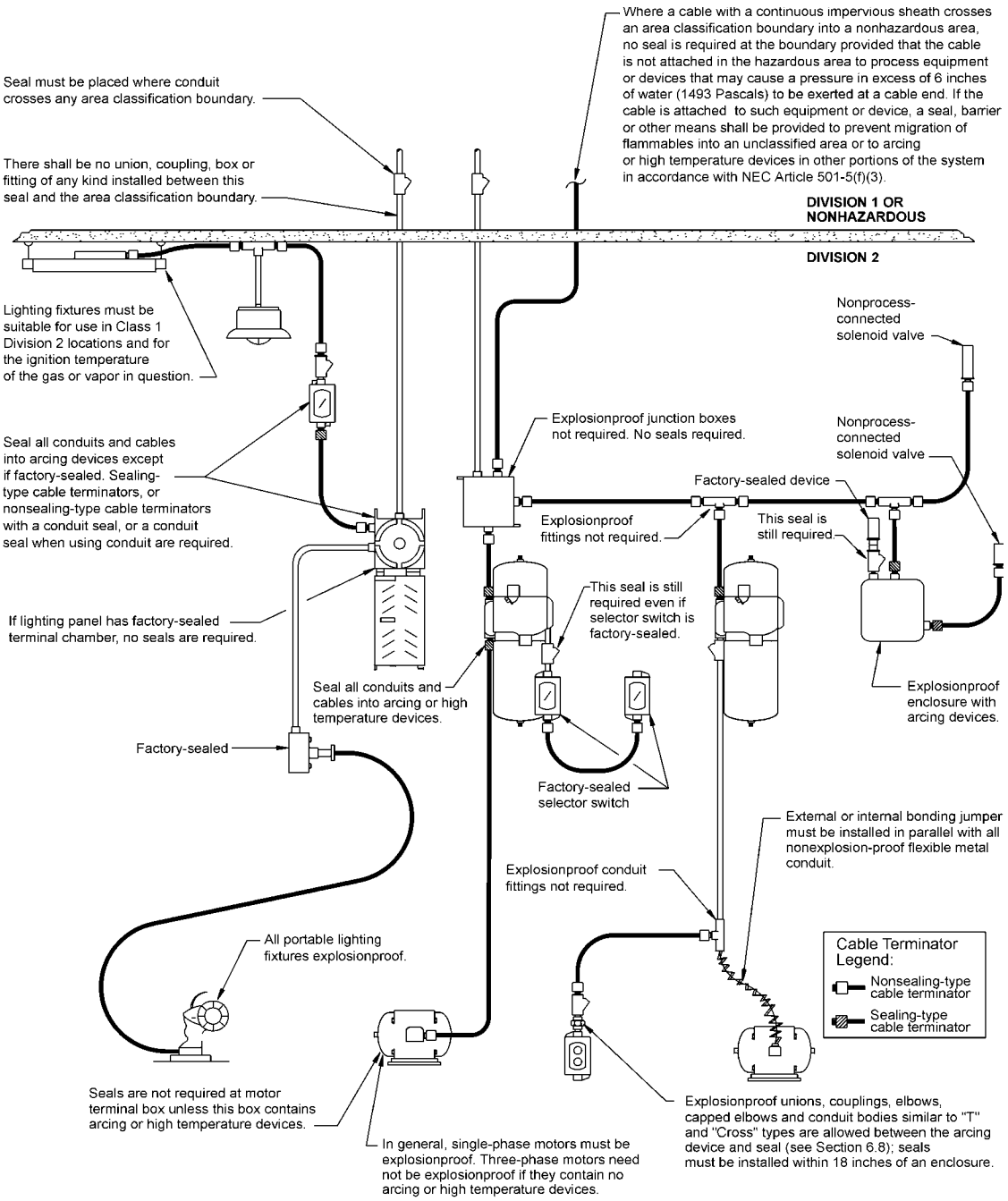
Figure 14 depicts a typical United States and Canadian Class I, Division 1 cable system installation.



Note: See Section 6 for cables approved for use in Div. 1 locations.

Figure 14 — Typical United States and Canadian Class I, Division 1 cable system installation (Reference API RP 14F, Figure 2, Recommended Practice for Design and Installation of Electrical Systems for Fixed and Floating Offshore Petroleum Facilities for Unclassified and Class I, Division 1 and Division 2 Locations)

Figure 15 depicts a typical United States and Canadian Class I, Division 2 conduit/cable system installation.



Note: See Section 6 for cables approved for use in Div. 2 locations.

Figure 15 — Typical United States and Canadian Class I, Division 2 conduit/cable system (Reference API RP 14F, Figure 3, Recommended Practice for Design and Installation of Electrical Systems for Fixed and Floating Offshore Petroleum Facilities for Unclassified and Class I, Division 1 and Division 2 Locations)

6.5 Comparisons of wiring methods [see Tables 4a and 4b]

6.5.1 Comparison of cable and conduit systems

Metallic conduit systems have been widely used in the United States and Canada to provide physical and environmental protection to conductors for both hazardous and nonhazardous locations. Conduit systems also prevent the transmission of an internally generated cable or conductor insulation fire within one conduit from spreading to an adjacent conduit, thus offering superior fire resistance. Ferrous metallic conduit can provide substantial magnetic shielding on sensitive circuits. Because conduit systems provide a closed system for the transmission of gases from one area to another, they must rely on properly located and properly installed conduit seals to provide explosionproof system integrity. Conduit systems are subject to internal condensation and resulting corrosion, which, if unchecked, can compromise explosion protection, especially with regard to offshore or shoreline locations. Also, under certain conditions, conduit systems can provide a passageway for liquids and gases, possibly transmitting corrosive or toxic substances to control rooms and electrical equipment buildings. In some countries, conduit systems are not recognized or accepted.

In many instances, cable systems offer an attractive alternative to conduit systems and are used worldwide. Cable systems may be less labor intensive during the initial installation and in follow-up maintenance. Certain types of cable construction and termination methods offer greater corrosion protection than conduit systems and offer an overall increase in system integrity due to reduced long-term corrosion effects. Cable systems provide the benefit that they are visible, which allows for ongoing inspection and simplifies the task of tracing circuits. There are numerous types of cables that are permitted in hazardous areas.

6.5.2 Comparisons between direct and indirect cable entries

A comparison between cable systems with direct and indirect entry shows some advantages to the indirect method, the system with terminal chambers that uses the INCREASED SAFETY type of protection. The installer can connect the indirect-entry type without opening the flameproof equipment chamber. With the cable system that uses direct entry, the flameproof protection completion can be achieved only during installation (on site).

Direct entry technology requires that the wiring connections be accomplished in the flameproof chamber. When using indirect entry, the connection is to terminals made in a separate terminal chamber that may be type of protection INCREASED SAFETY or FLAMEPROOF.

7 Grounding and bonding practices

7.1 In the United States and Canada, grounding and bonding practices in hazardous (classified) locations must follow the same standards as grounding and bonding practices in nonhazardous (unclassified) locations. In addition, the following special precautions must be followed:

a) Locknut bushings and double-locknut bushings must not be used as the only bonding method but must be paralleled with bonding jumpers. Reference NEC 501-16(a).

b) Flexible-metal conduit or liquid-tight flexible-metal conduit must not be used as the only grounding path but must be paralleled with internal or external bonding jumpers. Reference NEC 501-16(b).

c) All conduit must be threaded (NPT standard threads with 3/4 inch taper per foot) and made wrench-tight to prevent sparking when fault current flows through the conduit system. Reference NEC 500-2.

d) When required by the control drawing, intrinsically safe systems must be provided with a dedicated grounding conductor separate from the power system so that ground currents will not normally flow.

The systems must be reliably connected to a ground electrode in accordance with NEC Article 250 or CEC Part 1, Section 10. (Reference ISA-RP12.6)

7.2 Internationally, the term EARTHED is used instead of GROUNDED, but the same basic practices are followed.

Table 4a — Field wiring in United States Class I locations [a,b]

	Zone 0		Division 1 / Zone 1		Division 2 / Zone 2	
Wiring system	**IS**	**NIS**	**IS**	**NIS**	**IS or Nonincendive**	**NIS**
Threaded rigid metal conduit	A	X	A	A	A	A
Threaded steel intermediate metal conduit	A	X	A	A	A	A
Flexible metal explosionproof fitting	A	X	A	A	A	A
Types MI, MC, & ITC cable	A	X	A	A[dg]	A	A
Types PLTC, HV, ITC, and TC cable	A	X	A	X	A	A
Flexible metal conduit	A	X	A	X	A	A[c,f]
Liquid-tight, flexible metal conduit	A	X	A	X	A	A[c,f]
Electrical metallic tubing (steel)	A	X	A	X	A	X
Flexible cord	A	X	A	Note 1	A	A[c,e] Note 1 or 2
Any other wiring method suitable for nonhazardous locations	A	X	A	X	A	X

a Abbreviations: IS = intrinsically safe; NIS = not intrinsically safe; A = acceptable; X = not acceptable.
b See the NEC for description and use of wiring systems.
c Acceptable only where flexibility is needed.
d Acceptable only with termination fittings approved for Class I, Division 1 locations of the proper groups.
e Extra-hard usage type with a grounding conductor only acceptable.
f Special bonding/grounding methods are required.
g Type MC or Type ITC cable approved for Division 1 permitted in industrial establishments, see NEC 501-4 (a) Exception 2 or 3.

NOTE 1 — Acceptable on approved portable equipment where provisions are made for cord replacement, per NEC 501-11.

NOTE 2 — Acceptable on process control instruments to facilitate replacements, per NEC 501-3(b) (6).

Table 4b — Field wiring in United States Class II locations [a,b]

	Division 1		Division 2	
Wiring system	**IS**	**NIS**	**IS or Nonincendive**	**NIS**
Threaded rigid metal conduit	A	A	A	A
Threaded steel intermediate metal conduit	A	A	A	A
Flexible metal explosionproof fitting	A	A[c]	A	A[c]
Type MI cable	A[d]	A[e]	A	A
Type MC cable	A[d]	X[i]	A	A
Type PLTC, ITC, and TC cable	A[d]	X	A	A[f]
Flexible metal conduit	A[d]	X	A	X
Liquid-tight, flexible metal conduit	A[d]	A[c,d,g]	A	A[c,d,g]
Flexible cord	A[d]	A[c,d,h]	A	A[c,d,h]
Dust-tight wireways and raceways	A	X	A	A
Any other wiring method suitable for nonhazardous locations	A[d]	X	A	X
Electrically metallic tubing	A[d]	X	A	A

a Abbreviations: IS = intrinsically safe; NIS = not intrinsically safe; A = acceptable; X = not acceptable.
b See the NEC for description and use of wiring systems.
c Acceptable only where flexibility is needed.
d Acceptable only with dust-tight seals at both ends when electrically conductive dusts will be present.
e Acceptable only with termination fittings approved for Class I, Division 1 locations of the proper groups.
f Acceptable in ventilated channel-type cable trays in a single layer for a space not less than the larger cable diameter between adjacent cables.
g Special bonding methods are required.
h Only extra-hard usage type with a grounding conductor acceptable.
i Type MC cable approved for Division 1 permitted in industrial establishments, see NEC 501-4 (a) Exception 2.

8 Maintenance practices

Special attention must be focused on hazardous location equipment maintenance procedures in order to maintain the integrity of the original installation. The following are pertinent maintenance practices.

8.1 Hazardous location equipment can be repaired only in accordance with the manufacturer's instructions. Some codes of practice require the recording of repairs and the inspection of the completed repair by a second competent person.

8.2 Maintenance personnel should ensure that all explosionproof enclosures are properly closed and furnished with the proper number and type of fasteners. Care should be exercised to assure that all machine-finished flanges are protected from damage during maintenance to ensure surface integrity.

8.3 Maintenance personnel should ensure that all grounding conductors are properly terminated.

8.4 Any physical abnormalities noted should be corrected or reported to the next level of supervision.

8.5 All threaded connections (including flame paths) should be regularly coated with a lubricant approved for the purpose.

8.6 Defective circuit protection devices (primarily fuses) must be replaced with functional equivalent devices (proper amperage, voltage, characteristics, etc.).

8.7 Periodic inspections should be made to ensure that intrinsically safe circuits are isolated from non-intrinsically safe circuits.

8.8 Periodic inspections should be made to ensure that the equipment is suitable for the current area classification.

8.9 Special care must be taken to ensure that different intrinsically safe circuits do not become shorted together during calibration and maintenance.

Annex A — Acronyms

Throughout the text, many acronyms or abbreviations are used. The following list of acronyms provides a ready reference.

AIT	Autoignition Temperature
ANCE	Asociación Mexicana Nacional de Normalización y Certificación del Sector Electrico (Mexico)
ANSI	American National Standards Institute
API	American Petroleum Institute
AS	Australian Standard
ASTM	American Society for Testing and Materials
AWG	American Wire Gauge
BASEEFA	British Approvals Service for Electrical Equipment in Flammable Atmospheres - Part of EECS (UK)
BS	British Standard
BSI	British Standards Institute
BVFA	Bundesversuchs - und Forschungs - Anstalt (Austria)
BvS	Berggewerkschafitliche Versuchsstrecke [see DMT] (Germany)
CANMET	Canadian Mining and Energy Technology (Canada)
CEC	Canadian Electrical Code
CEN	European Committee for Standardization
CENELEC	European Committee for Electrotechnical Standardization
CEPEL	Centro de Pesquisas de Energia Electrica (Brazil)
CERCHAR	Centre d'Etudes et Recherches des Chourbonnage de France [See INERIS] (France)
CESI	Centro Elettrotecnico Sperimentale Italiano (Italy)
CSA	Canadian Standards Association (Canada)
CSIR	Central Mining Research Institute (India)
DEMKO	Danmarks Elektriske Material Kontrol A/S (Denmark)
DIN	Deutsche Institut fur Normung e.V. (Germany)

DIP	Dust-ignitionproof
DNV	Det Norske Veritas (Norway)
DMT	Deutsche MontanTechnologie [See BvS] (Germany)
EECS	Electrical Equipment Certification Service (UK)
EMR	Energy Mines and Resources (Canada)
EN	European Norm (Standard)
EP	Explosionproof
ERA	Electrical Research Association (UK)
FM	Factory Mutual Research Corporation (USA)
GOST	Gossudarstwenny Standard (Russia)
HSE	Health and Safety Executive (UK)
IEC	International Electrotechnical Commission
IECEE	International Electrotechnical Commission Committee on Electrical Equipment
IEEE	Institute of Electrical and Electronics Engineers
IEV	International Electrotechnical Vocabulary (IEC 50)
INIEX	Institut National des Industries Extractives [See ISSeP] (Belgium)
INERIS	Institut National de L'Environnement Industriel et des Risques (France)
INSEMEX	Institutul National Pentru Securitate Miniera si Protectie Antiexploziva (Romania)
IP	Institute of Petroleum (UK)
IP	Ingress Protection per IEC 529
ISA	Instrument Society of America
ISSeP	Institute Scientific des Service Public [was INIEX] (Belgium)
ISO	International Organization for Standardization
ITS	Intertek Testing Services
JIS	Japan Industry Standards (Japan)
KEMA	Keuring van Elektrotechnische Materialen (The Netherlands)
LCIE	Laboratoire Central Des Industries Electriques (France)

LEL	Lower Explosive Limit (Lower Flammable Limit)
LFL	Lower Flammable Limit (Lower Explosive Limit)
LOM	Laboratorio Oficial Madariaga (Spain)
LOSC	Londonderry Occupational Safety Centre—Work Cover Authority (Australia)
MECS	Mining Equipment Certification Service—Part of EECS (UK)
MESG	Maximum Experimental Safe Gap
MIC	Minimum Igniting Current
MIE	Minimum Ignition Energy
MMS	Minerals Management Service, U.S. Department of the Interior
MSHA	Mine Safety and Health Administration
NAMUR	Normenarbeitsgemeinschaft fur Mess- und Regelungstechnik in der Chemischen Industrie (German)
NAS	National Academy of Science
NEC	National Electrical Code (ANSI/NFPA 70)
NEMA	National Electrical Manufacturers Association
NEMKO	Norges Elektriske Materiellkontroll (Norway)
NEPSI	National Supervision and Inspection Centre for Explosion Protection and Safety of Instrumentation (China)
NFPA	National Fire Protection Association
NRC	National Research Council
NRTL	Nationally Recognized Testing Laboratory
NMX	Voluntary Standards (Mexico)
NOM	Norma Official Mexicana (Mexico)
OSHA	Occupational Safety and Health Act (or Administration)
PTB	Physikalisch-Technische Bundesanstalt (Germany)
RIIS	The Research Institute of Industrial Safety of the Ministry of Labour (Japan)
SAA	Standards Association of Australia
SABS	South African Bureau of Standards (South Africa)

SCS	Sira Certification Service (UK)
SEV	Schweizerischer Elektrotechnischer Verein (Switzerland)
SIMTARS	Safety in Mines Testing and Research Station (Australia)
SIPAI	Shanghai Institute of Process Automation Instrumentation (China)
SNZ	Standards New Zealand
SMRE	Safety in Mines Research Establishment - Operation Suspended (UK)
SP	Sveriges Provnings – och Forskningsinstitut trans: Swedish National Testing and Research Institute (Sweden)
TIIS	Technical Institute of Industrial Safety (Japan)
TUV-A	Technischer Uberwachungsverein Austria (Austria)
UEL	Upper Explosive Limit (Upper Flammable Limit)
UFL	Upper Flammable Limit (Upper Explosive Limit)
UK	United Kingdom
UL	Underwriters Laboratories Inc. (USA)
ULC	Underwriters' Laboratories of Canada
USA	United States of America
USCG	United States Coast Guard
USBM	*Former* United States Bureau of Mines
VTT	Vaition Tekmillinen Tutkimuslaitos (Finland)

Annex B — References

AMERICAN GAS ASSOCIATION

AGA XF0277 Classification of Gas Utility Areas

Available from:

AGA	American Gas Association 1515 Wilson Boulevard Arlington, VA 22209	Tel: 703-841-8564

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C34.1 Voltage Ratings for Electrical Power Systems and Equipment (60 Hz)

Available from:

ANSI	American National Standards Institute 1 West 42nd Street New York, NY 10036	Tel.: (212) 642-4900 Fax: (212) 302-1286 http://www.ansi.org

AMERICAN PETROLEUM INSTITUTE (API)

RP 14F Recommended Practice for Design and Installation of Electrical Systems for Fixed and Floating Offshore Production Facilities for Unclassified and Class I, Division 1 and Division 2 Locations

RP 14FZ Recommended Practice for Design and Installation of Electrical Systems for Fixed and Floating Offshore Production Facilities for Unclassified and Class I, Zone 0, Zone 1, and Zone 2 Locations

RP 500 Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Division 1 and Division 2

RP 505 Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1, and Zone 2.

Available from:

API	American Petroleum Institute 1220 L Street, NW Washington, DC 20005-4070	Tel.: (202) 682-8000 Fax: (202) 682-8051

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS, INC. (ASHRAE)

ASHRAE Fundamentals Handbook

Available from:

ASHRAE American Society Of Heating, Refrigerating And Air Conditioning Engineers
1791 Tullie Circle, N.E.
Atlanta, GA 30329

EUROPEAN COMMITTEE FOR ELECTROTECHNICAL STANDARDISATION (CENELEC)

EN 50 014, General Requirements

EN 50 015, Oil Immersion ‘o’

EN 50 016, Pressurized Apparatus ‘p’

EN 50 017, Powder Filling ‘q’

EN 50 018, Flameproof Enclosure ‘d’

EN 50 019, Increased Safety ‘e’

EN 50 020, Intrinsic Safety ‘i’

prEN 50 021, Type of Protection ‘n’ (Draft)

EN 50 028, Encapsulation ‘m’

EN 50 039, Intrinsically Safe Electrical Systems ‘i’

EN 50 054, 055, 056, 057, and 058; Instruments for the Detection of Combustible Gases

Available from:

ANSI American National Standards Institute
11 West 42nd Street
New York, NY 10036
Tel.: (212) 642-4900
Fax: (212) 302-1286
http://www.ansi.org

or

BSI British Standards Institute
389 Chiswick High Road
London, W4 4AL
Tel.: 44-181-996-7000
Fax: 44-181-996-7001

or

CENELEC European Committee for Electrotechnical Standardization
Central Secretariat
rue de Stassart, 35
B-1050 Brussels
Belgium

CANADIAN STANDARDS ASSOCIATION (CSA)

C22.1, Part 1	Canadian Electrical Code
C22.2, No. 25	Enclosures for Class II, Groups E, F, and G Hazardous Locations
C22.2, No. 30	Explosionproof Enclosures for Use in Class I Hazardous Locations
C22.2, No. 145	Motors and Generators for Use in Hazardous Locations
C22.2, No. 152	Combustible Gas Detection Equipment
C22.2, No. 157	Intrinsically Safe and Nonincendive Equipment for Use in Hazardous Locations
C22.2, No. 174	Cables and Cable Glands for Use in Hazardous Locations
C22.2, No. 213	Nonincendive Electrical Equipment for Use in Class I, Division 2 Hazardous Locations
ISBN 0-921347-39-1	Guide for the Design, Testing, Construction, and Installation of Equipment in Explosive Atmospheres (2nd Edition), John A. Bossert

Available from:

CSA Canadian Standards Association
178 Rexdale Boulevard
Etobicoke, Ontario M9W 1R3
Canada
Tel.: (416) 747-4000
Fax: (416) 747-4178
http://www.csa.ca

FACTORY MUTUAL RESEARCH CORPORATION (FM)

Approval Standard Class No. 3600	Electrical Equipment for Use in Hazardous (Classified) Locations, General Requirements
Approval Standard Class No. 3610	Intrinsically Safe Apparatus and Associated Apparatus for Use in Class I, II, and III, Division 1 Hazardous Locations
Approval Standard Class No. 3611	Electrical Equipment for Use in Class I, Division 2, Class II, Division 2 and Class III, Divisions 1 and 2 Hazardous Locations
Approval Standard Class No. 3615	Explosionproof Electrical Equipment
Approval Standard Class No. 3620	Purged and Pressurized Electrical Equipment for Hazardous (Classified) Locations
Approval Standard Class Nos. 6310-6330	Combustible Gas Detectors

Available from:

FMRC	Factory Mutual Research Corporation 1151 Boston-Providence Turnpike Norwood, MA 02062	Tel.: (781) 762-4300 Fax: (781) 762-9375 http://www.factorymutual.com

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

Std. 45	Recommended Practice for Electrical Installation on Shipboard
Std. 142	Recommended Practice for Grounding of Industrial and Commercial Power Systems

Available from:

IEEE	Institute of Electrical and Electronic Engineers 445 Hoes Lane P.O. Box 1331 Piscataway, NJ 08855-1331	Tel: (800) 678-4333 Fax: (732) 562-5445

INSTITUTE OF PETROLEUM (IP)

IP - Part 15	Model Code of Safe Practice in the Petroleum Industry, Part 15: Area Classification Code for Petroleum Installations

Available from:

IP	Institute of Petroleum 61 New Cavendish Street London, UK W1M 8AR

ISA

ISA-S5.1	Instrumentation Symbols and Identification
ISA-S7.3	Quality Standard for Instrument Air
ISA-S12.0.01 (IEC 79-0 Mod)	Electrical Apparatus for Use in Class I, Zones 0 and 1, Hazardous (Classified) Locations: General Requirements
ISA-S12.01.01	Definitions and Information Pertaining to Electrical Apparatus in Hazardous (Classified) Locations
ISA-dS12.2.01 (IEC 79-11 Mod)	Electrical Apparatus for use in Class I, Zones 0, 1, and 2 Hazardous (Classified) Locations: Type of Protection — Intrinsic Safety “i”
ISA-RP12.2.02	Recommendations for the Preparation, Content, and Organization of Intrinsic Safety Control Drawings
ISA-RP12.4	Pressurized Enclosures
ISA-RP12.6	Wiring Practices for Hazardous (Classified) Locations Instrumentation Part I: Intrinsic Safety

ISA-S12.10	Area Classification in Hazardous (Classified) Dust Locations
ISA-S12.11	Electrical Equipment in Hazardous Dust Locations
ISA-S12.12	Electrical Equipment for Use in Class 1, Division 2 Hazardous (Classified) Locations
ISA-dS12.12.01 (IEC 79-15 Mod)	Electrical Apparatus for Use in Class I, Zone 2 Hazardous (Classified) Locations: Type of Protection — “n”
ISA-S12.13, Part I	Performance Requirements, Combustible Gas Detectors
ISA-S12.13, Part II	Installation, Operation, and Maintenance of Combustible Gas Detection Instruments
ISA-S12.15, Part I	Performance Requirements for Hydrogen Sulfide Detection Instruments (10-100 ppm)
ISA-RP12.15, Part II	Installation, Operation, and Maintenance of Hydrogen Sulfide Detection Instruments
ISA-S12.16.01 (IEC 79-7 Mod)	Electrical Apparatus for Use in Class I, Zones 1 and 2 Hazardous (Classified) Locations: Type of Protection — Increased Safety “e”
ISA-S12.22.01 (IEC 79-1 Mod)	Electrical Apparatus for Use in Class I, Zone 1 Hazardous (Classified) Locations: Type of Protection — Flameproof “d”
ISA-S12.23.01 (IEC 79-18 Mod)	Electrical Apparatus for Use in Class I, Zone 1 Hazardous (Classified) Locations: Type of Protection — Encapsulation “m”
ISA-RP12.24.01 (IEC 79-10 Mod)	Recommended Practice for Classification of Locations for Electrical Installations Classified as Class I, Zone 0, Zone 1, or Zone 2
ISA-S12.25.01 (IEC 79-5 Mod)	Electrical Apparatus for Use in Class I, Zones 1 and 2 Hazardous (Classified) Locations: Type of Protection — Powder Filling “q”
ISA-S12.26.01 (IEC 79-6 Mod)	Electrical Apparatus for Use in Class I, Zones 1 and 2 Hazardous (Classified) Locations: Type of Protection — Oil Immersion “o”
ISA-S51.1	Process Instrumentation Terminology
ISA-S71.01	Environmental Conditions for Process Measurement and Control Systems: Temperature and Humidity
ISA-S82.01	Safety Standard for Electrical and Electronic Test, Measuring, Controlling, and Related Equipment — General Requirements
ISA-S82.02	Safety Standard for Electrical and Electronic Test, Measuring, Controlling, and Related Equipment — Electrical and Electronic Test and Measuring Equipment
ISA-S82.03	Safety Standard for Electrical and Electronic Test, Measuring, Controlling, and Related Equipment — Electrical and Electronic Process Measurement and Control Equipment

Calder, W. and Magison, E.C., Electrical Safety in Hazardous Locations, ISA, 1983.

Magison, E.C., Electrical Instruments in Hazardous Locations, ISA, 4th Edition - 1998.

Available from:

ISA	ISA 67 Alexander Drive, P.O. Box 12277 Research Triangle Park, NC 27709	Tel.: (919) 549-8411 Fax: (919) 549-8288 http://www.isa.org

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

60050	International Electrotechnical Vocabulary (IEV)
60079-0	Electrical Apparatus for Explosive Gas Atmospheres: General Requirements
60079-1	Construction and Verification Test of Flameproof Enclosures of Electrical Apparatus, Amendment No. 1, Type of Protection ‘d’
60079-1A	Construction and Verification Test of Flameproof Enclosures of Electrical Apparatus — First Supplement: Appendix D: Method of Test for Ascertainment of Maximum Experimental Safe Gap
60079-2	Electrical Apparatus (pressurization), Type of Protection ‘p’
60079-3	Spark Test Apparatus for Intrinsically Safe Circuits
60079-4	Method of Test for Ignition Temperature
60079-4A	Method of Test for Ignition Temperature, First Supplement
60079-5	Powder-Filling, Type of Protection ‘q’
60079-6	Oil-Immersion, Type of Protection ‘o’
60079-7	Increased Safety, Type of Protection ‘e’
60079-10	Classification of Hazardous Areas
60079-11	Intrinsic Safety, Type of Protection ‘i’
60079-12	Classification of Mixtures of Gases or Vapors with Air According to Their Maximum Experimental Safe Gaps and Minimum Igniting Currents
60079-13	Construction and Use of Rooms and Buildings Protected by Pressurization
60079-14	Electrical Installation in Explosive Gas Atmospheres (Other than Mines)
60079-15	Electrical Apparatus, Type of Protection ‘n’
60079-16	Artificial ventilation for the protection of analyzer houses

60079-17	Inspection and maintenance of electrical installations in hazardous areas (other than mines)
60079-18	Encapsulation, Type of Protection ‘m’
60079-19	Repair and overhaul for apparatus used in explosive atmospheres (other than mines or explosives)
60079-20	Data for flammable gases and vapours relating to the use of electrical apparatus
60529	Degrees of protection provided by enclosures (IP Code)
60654-1	Temperature, Humidity, and Barometric Pressure
61241	Electrical Apparatus for use in the presence of combustible dust
61241-1-1	Part 1: Electrical apparatus protected by enclosures — Section 1: Specification for apparatus
61241-1-2	Part 1: Electrical apparatus protected by enclosures — Section 2: Selection, installation, and maintenance
61241-2-1	Part 2: Test methods — Section 1: Methods for determining the minimum ignition temperatures of dust
61241-2-2	Part 2: Test methods — Section 2: Method for determining the electrical resistivity of dust in layers
61241-2-3	Part 2: Test methods — Section 3: Method for determining minimum ignition energy of dust/air mixtures
61241-3	Part 3: Classification of areas where combustible dusts are or may be present.
61892-7	Mobile and Fixed Offshore Units — Electrical Installations Part 7: Hazardous Areas

Available from:

ANSI	American National Standards Institute 11 West 42nd Street New York, NY 10036	Tel.: (212) 642-4900 Fax:(212) 302-1286 http://www.ansi.org
or		
IEC	International Electrotechnical Commission Bureau Centrale de la Commission Electrotechnique International 3, rue de Varembe Case postale 131, 1211 Geneva 20, Switzerland	Tel: 41-22-734-01-50 Tel: 41-22-734-01-50 Fax:41-22-733-38-43 http://www.iec.ch

MISCELLANEOUS

Hilado, C.J., "A Method for Estimating Limits of Flammability," Journal Fire Flammability, Vol. 6, pp. 130-139 (April 1975).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 6 Enclosures for Industrial Controls and Systems

No. 250 Enclosures for Electrical Equipment

Available from:

NEMA National Electrical Manufacturers Association
1300 North 17th Street
Suite 1847
Rosslyn, VA 22209
Tel.: (703) 841-3200
Fax: (703) 841-3300
http://www.nema.org

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 30 Flammable and Combustible Liquids Code

No. 37 Installation and Use of Stationary Combustion Engines and Turbines

No. 45 Fire Protection for Laboratories Using Chemicals

No. 70 National Electrical Code

No. 77 Recommended Practice on Static Electricity

No. 90A Installation of Air Conditioning and Ventilating Systems

No. 91 Installation of Blower and Exhaust Systems for Dust, Stock, and Vapor Removal or Conveying

No. 325 Guide to Fire Hazard Properties of Flammable Liquids, Gases, and Volatile Solids

No. 496 Purged and Pressurized Enclosures for Electrical Equipment in Hazardous (Classified) Locations

No. 497 Recommended Practice on Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installation in Chemical Plants

No. 499 Classification of Combustible Dusts and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas

Available from:

NFPA National Fire Protection Association
P.O. Box 9146
Quincy, MA 02269-9959
Tel.: (617) 770-3000
Fax: (617) 770-0700

UNDERWRITERS LABORATORIES INC. (UL)

UL Technical Report No. 58	An Investigation of Flammable Gases or Vapors with Respect to Explosionproof Electrical Equipment
UL 583	Standard for Electric-Battery-Powered Industrial Trucks
UL 674	Standard for Electric Motors and Generators for Use in Hazardous (Classified) Locations, Class I, Groups C and D, and Class II, Groups E, F, and G
UL 698	Standard for Industrial Control Equipment for Use in Hazardous (Classified) Locations
UL 781	Standard for Portable Electric Lighting Units for Use in Hazardous (Classified) Locations
UL 783	Standard for Electric Flashlights and Lanterns for Use in Hazardous (Classified) Locations
UL 823	Standard for Electric Heaters for Use in Hazardous (Classified) Locations
UL 844	Standard for Electric Lighting Fixtures for Use in Hazardous (Classified) Locations
UL 877	Standard for Circuit Breakers and Circuit-Breaker Enclosures for Use in Hazardous (Classified) Locations, Class I, Groups A, B, C, and D, and Class II, Groups E, F, and G
UL 886	Standard for Outlet Boxes and Fittings for Use in Hazardous (Classified) Locations
UL 894	Standard for Switches for Use in Hazardous (Classified) Locations
UL 913	Standard for Intrinsically Safe Apparatus and Associated Apparatus for Use in Class I, II, and III, Division 1, Hazardous (Classified) Locations
UL 1002	Standard for Electrically Operated Valves for Use in Hazardous (Classified) Locations, Class I, Groups A, B, C, and D, Class II, Groups E, F, and G
UL 1010	Standard for Receptacle-Plug Combinations for Use in Hazardous (Classified) Locations
UL 1067	Standard for Electrically Conductive Equipment and Materials for Use in Flammable Anesthetizing Locations
UL 1203	Explosionproof and Dust-Ignitionproof Electrical Equipment for Use in Hazardous (Classified) Locations
UL 1207	Standard for Sewage Pumps for Use in Hazardous (Classified) Locations
UL 1604	Electrical Equipment for Use in Hazardous (Classified) Locations, Class I and II, Division 2, and Class III, Divisions 1 and 2 Hazardous (Classified) Locations
UL 2225	Standard for Metal-Clad Cables and Cable-Sealing Fittings for Use in Hazardous (Classified) Locations

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Englewood, CO 80112

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Underwriters Laboratories
333 Pfingston Road
Northbrook, IL 60062-2096

Tel.:(847) 272-8800
Fax: (847) 272-8125
http://www.ul.com

UNITED STATES CODE OF FEDERAL REGULATIONS

Title 30, Parts 1 through 199, Mineral Resources

Title 30, Part 250, Oil and Gas and Sulfur Operations in the Outer Continental Shelf

Title 46, Parts 110 through 113, Shipping Subchapter J, Electrical Engineering, (United States Coast Guard)

Available from:

Superintendent of Documents
U.S. Government Printing Office
Washington, D.C. 20402

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Fax: (303) 397-7935
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Annex C — Listing of worldwide-codes, guides, and standards

NOTE — Annex C is filled as much as possible at the time of publication, but should not be considered complete. Additions known to the users for consideration for future editions should be directed to:

ISA
67 Alexander Drive
P.O. Box 12277
Research Triangle Park, NC 27709
Attn: Standards Department/SP12

Tel: 919-549-8411
Fax: 919-549-8288
standards@isa.org

Country	Australia	Austria*	Belgium*	Canada	CENELEC
Standards	**SAA**	**ETVA**	**ISSeP**	**CSA**	
General Requirements	AS 2380.1 AS 2381 AS 3000 Section 9	OVE-EX ENx 50 014 OVE-EX 65	NBN C23-001 EN 50 014	CEC 22.1 CSA E79-0	EN 50 014 (1st Ed) EN 50 014 (2nd Ed)
Oil Immersion/Type ‘o’		OVE-EX EN 50 015	NBN C23-104 EN 50 015	CSA E79-6	EN 50 015 (1st Ed) EN 50 015 (2nd Ed)
Pressurization/Type ‘p’	AS 2380.4	OVE-EX EN 50 016	NBN C23-105 EN 50 016	NFPA 496 CSA E79-2	EN 50 016 (1st Ed) EN 50 016 (2nd Ed)
Powder Filling/Type ‘q’		IEC 60079-5	NBN C23-106 EN 50 017	CSA E79-5	EN 50 017 (1st Ed) EN 50 017 (2nd Ed)
Explosionproof/Flameproof/Type ‘d’	AS 2380.2	OVE-EX EN 50 018	NBN C23-103 EN 50 018	C22.2 No. 30 CSA E79-1	EN 50 018 (1st Ed) EN 50 018 (2nd Ed)
Increased Safety/Type ‘e’	AS 2380.6	OVE-EX EN 50 019	NBN C23-102 EN 50 019	CSA E79-7	EN 50 019 (1st Ed) EN 50 019 (2nd Ed)
Intrinsic Safety/Type ‘i’	AS 2380.7	OVE-EX EN 50 020	NBN C23-101 EN 50 020	C22.2 No. 157 CSA E79-11	EN 50 020 (1st Ed) EN 50 020 (2nd Ed)
Type ‘n’	AS 2380.9			C22.2 No. 213 CSA E79-15	prEN 50 021
Encapsulation/Type ‘m’	AS 2431	IEC 60079-18		CSA E79-18	EN 50 028
Intrinsically Safe Electrical Systems	AS 2380.7		RGPT Article 251 bis NBN C23-201 EN 50 039	CEC Annex F	EN 50 039
Classification of Hazardous Gas & Vapor Locations	AS 2430 Part 1	OVE-EX65 EX-RL Guidelines No. 11	NBN C23-001 EN 50 014	CEC Section 18	EN 60 079-10
Classification of Mixtures of Gases & Vapors			NBN C23-001 EN 50 014	CEC Annex B	EN 50 014 (1st Ed) EN 50 014 (2nd Ed) prEN 60 079-20
Construction of room & bldg. protected by pressurization	AS 2380.4			IEC E79-13	
Area Classification in Hazardous Dust Locations	AS 2430 Part 2			CEC Section 18	prEN 6 1241-3
Electrical Instruments in Hazardous Dust Locations	AS 2236	OVE A50		C22.2 No. 25	prEN 6 1241-1-1
*CENELEC member country					

Country	Denmark*	Finland*	France*	Germany*
Standards	**DEMKO**		**UTE**	**VDE**
General Requirements	Afsnit 7A Afsnit 50 EN 50 014	SFS 4094 EN 50 014	NF C23-514 NF EN 50 014	Elexv; VDE 0165; EX-RL VDE 0170/0171 Part 1 DIN EN 50 014
Oil Immersion/Type ‘o’	Afsnit 50-1 EN 50 015	SFS 4095 EN 50 015	NF C23-515 NF EN 50 015	VDE 0170/0171 Part 2 DIN EN 50 015
Pressurization/Type ‘p’	Afsnit 50-2 EN 50 016	SFS 4096 EN 50 016	NF C23-516 NF EN 50 016	VDE 0170/0171 Part 3 DIN EN 50 016
Powder Filling/Type ‘q’	Afsnit 50-3 EN 50 017	SFS 4097 EN 50 017	NF C23-517 NF EN 50 017	VDE 0170/0171 Part 4 DIN EN 50 017
Explosionproof/Flameproof/Type ‘d’	Afsnit 50-4 EN 50 018	SFS 4098 EN 50 018	NF C23-518 NF EN 50 018	VDE 0170/0171 Part 5 DIN EN 50 018
Increased Safety/Type ‘e’	Afsnit 50-5 EN 50 019	SFS 4099 EN 50 019	NF C23-519 NF EN 50 019	VDE 0170/0171 Part 6 DIN EN 50 019
Intrinsic Safety/Type ‘i’	Afsnit 50-6 EN 50 020	SFS 4100 EN 50 020	NF C23-520 NF EN 50 020	VDE 0170/0171 Part 7 DIN EN 50 020
Type ‘n’				prEN 50 021
Encapsulation/Type ‘m’			NF C23-528 NF EN 50 028	VDE 0170/0171 Part 9 DIN EN 50 028
Intrinsically Safe Electrical Systems			NF C23-539 NF EN 50 039	VDE 0170/0171 Part 10 DIN EN 50 039
Classification of Hazardous Gas & Vapor Locations	Afsnit 7A	Section 41		VDE 0165 DIN EN 60 079-10
Classification of Mixtures of Gases & Vapors	Afsnit 7A			VDE 0165 VDE 0171
Construction of room & bldg. protected by pressurization				
Area Classification in Hazardous Dust Locations	Afsnit 7A			
Electrical Instruments in Hazardous Dust Locations		SFS 2972	NF C20-010	DIN 40050
*CENELEC member country				

Country	Greece*	Hungary	IEC	Irish Republic*	Italy*
Standards					**CEI**
General Requirements	IEC 60079-0	IEC 60079-0	IEC 60079-0	IS 231	CEI 31-8 CEI 64-2 EN 50 014
Oil Immersion/Type ‘o’	IEC 60079-6	IEC 60079-6	IEC 60079-6	IS 232	CEI 31-5 EN 50 015
Pressurization/Type ‘p’	IEC 60079-2	IEC 60079-2	IEC 60079-2	IS 233	CEI 31-2 EN 50 016
Powder Filling/Type ‘q’	IEC 60079-5	IEC 60079-5	IEC 60079-5	IS 234	CEI 31-6 EN 50 017
Explosionproof/Flameproof/Type ‘d’	IEC 60079-1	MS2 4814 IEC 60079-1	IEC 60079-1	IS 235	CEI 31-1 EN 50 018
Increased Safety/Type ‘e’	IEC 60079-7	IEC 60079-7	IEC 60079-7	IS 236	CEI 31-7 EN 50 019
Intrinsic Safety/Type ‘i’	IEC 60079-11	IEC 60079-11	IEC 60079-11	IS 237	CEI 31-9 EN 50 020
Type ‘n’	IEC 60079-15	IEC 60079-15	IEC 60079-15		CEI 31-11
Encapsulation/Type ‘m’	IEC 60079-18	IEC 60079-18	IEC 60079-18		CEI 31-13 EN 50 028
Intrinsically Safe Electrical Systems					
Classification of Hazardous Gas & Vapor Locations	IEC 60079-10	IEC 60079-10	IEC 60079-10		CEI 64-2
Classification of Mixtures of Gases & Vapors	IEC 60079-12	IEC 60079-12	IEC 60079-12		
Construction of room & bldg. protected by pressurization	IEC 60079-13	IEC 60079-13	IEC 60079-13		
Area Classification in Hazardous Dust Locations			IEC 61241-3		
Electrical Instruments in Hazardous Dust Locations					
*CENELEC member country					

Country	Japan	Netherlands*	Norway*	Poland	Portugal*
Standards	**JIS**	**NEN**	**NE**		
General Requirements	JIS C0903	NEN-EN 50 014	NEN-110 NVE Communication 1/77 NEK-EN 50 014		Decree No. 740/74 Decree No. 36270
Oil Immersion/Type ‘o’		NEN-EN 50 015	NEN-111 NEK-EN 50 015		
Pressurization/Type ‘p’	JIS C0903 JIS C0904	NEN-EN 50 016	NEN-112 NEK-EN 50 015		
Powder Filling/Type ‘q’		NEN-EN 50 017	NEN-113 NEK-EN 50 017		
Explosionproof/Flameproof/Type ‘d’	JIS C903 JIS C904	NEN-EN 50 018	NEN-114 NEK-EN 50 018	PN 721E	
Increased Safety/Type ‘e’	JIS C903 JIS C904 JIS C905	NEN-EN 50 019	NEN-115 NEK-EN 50 019		
Intrinsic Safety/Type ‘i’	JIS C903 JIS C904	NEN-EN 50 020	NEN-116 NEK-EN 50 020		
Type ‘n’		NEN-3125			
Encapsulation/Type ‘m’		NEN-EN 50 028	NEK-EN 50 028		
Intrinsically Safe Electrical Systems					
Classification of Hazardous Gas & Vapor Locations	BS 5345 Part 1	Directorate-General of Labor Report No. 2E	NVE Communication 1/77		
Classification of Mixtures of Gases & Vapors	VDE 0171 VDE 0165				
Construction of room & bldg. protected by pressurization					
Area Classification in Hazardous Dust Locations					
Electrical Instruments in Hazardous Dust Locations	Labour Ordinance No. 16 RIIS	NEN-1010	NVE Communication 1/77 Annex II		
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Country	Russia	S. Africa	Spain*	Sweden*	Switzerland*
Standards	**GOST**	**SABS**	**SIEV/SAT**	**UNE**	**SEV**
General Requirements		SABS 086 SABS 087 SABS 808 SABS IEC 60079-0	SS EN 50 014 SEN 421 0819	Instruccion MI.BT UNE 20318 SS EN 50 014	SEV 1068 SS EN 50 014
Oil Immersion/Type ‘o’		SABS IEC 60079-6	SS EN 50 015	UNE 20326 UNE 21815 SS EN 50 015	SEV 1069 SS EN 50 015
Pressurization/Type ‘p’		SABS 0119 SABS IEC 60079-2	SS EN 50 016	UNE 20319 UNE 21816 SS EN 50 016	SEV 1070 SS EN 50 016
Powder Filling/Type ‘q’		SABS IEC 60079-5	SS EN 50 017	UNE 20321 UNE 21817 SS EN 50 017	SEV 1071 SS EN 50 017
Explosionproof/Flameproof/Type ‘d’		SABS 314 SABS IEC 60079-1	SS EN 50 018	UNE 20320 UNE 21818 SS EN 50 018	SEV 1072 SS EN 50 018
Increased Safety/Type ‘e’		SABS 1031 SABS IEC 60079-7	SS EN 50 019	UNE 20327 UNE 20328 SS EN 50 019	SEV 1073 SS EN 50 019
Intrinsic Safety/Type ‘i’		SABS 549 SABS IEC 60079-11	SS EN 50 020 SEN 421 0879	UNE 20327 UNE 21820 SS EN 50 020	SEV 1074 SS EN 50 020
Type ‘n’		SABS 970			
Encapsulation/Type ‘m’					
Intrinsically Safe Electrical Systems			EN 50 039		SEV 3538
Classification of Hazardous Gas & Vapor Locations		SABS 089 Part II SABS 0108	SAI SIND-FS SS 421 0820 SS 421 0821	MI.BT 009 MI.BT 026 UNE 20322	SEV 3307-1
Classification of Mixtures of Gases & Vapors				UNE 20320	
Construction of room & bldg. protected by pressurization			SEN 421 0823		
Area Classification in Hazardous Dust Locations					
Electrical Instruments in Hazardous Dust Locations		SABS 969	SEN 2121 SS IEC 529		SEV 1000
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Country	U.K.*	U.S.A	U.S.A.	U.S.A	U.S.A.
Standards	**BS**	**API**	**FM**	**IEEE**	**ISA**
General Requirements	BS 5501 Part 1 EN 50 014 BS EN 50 014 IEC 60079-0		Class No. 3600 (Divisions) Class No. 3601 (Zones)		ISA-S12.0.01 (Zones)
Oil Immersion/Type ‘o’	BS 5501 Part 2 EN 50 015 BS EN 50 015 IEC 60079-6		Class No. 3621		ISA-S12.26.01
Pressurization/Type ‘p’	BS 5501 Part 3 EN 50 016 BS EN 50 016 IEC 60079-2		Class No. 3620		ISA-RP12.4
Powder Filling/Type ‘q’	BS 5501 Part 4 EN 50 017 BS EN 50 01 IEC 60079-5		Class No. 3622		ISA-S12.25.01
Explosionproof/Flameproof/Type ‘d’	BS 5501 Part 5 EN 50 018 BS EN 50 018 IEC 60079-1		Class No. 3615		ISA-S12.22.01
Increased Safety/Type ‘e’	BS 5501 Part 6 EN 50 019 BS EN 50 019 IEC 60079-7		Class No. 3619		ISA-S12.16.01
Intrinsic Safety/Type ‘i’	BS 5501 Part 7 EN 50 020 BS EN 50 020 IEC 60079-11		Class No. 3610		ISA-RP12.6
Type ‘n’	BS 4533 BS 5000 BS 6941 IEC 60079-15 BS/prEN 50 021		Class No. 3611	IEEE Std. 303	ISA-S12.12.01
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Country	U.K.*	U.S.A	U.S.A.	U.S.A	U.S.A.
Standards	**BS**	**API**	**FM**	**IEEE**	**ISA**
Encapsulation/Type ‘m’	BS 5501 Part 8 EN 50 028 BS EN 50 028 IEC 60079-18		Class No. 3614		ISA-S12.23.01
Intrinsically Safe Electrical Systems	BS 5501 Part 9 EN 50 039 BS EN 50 039		Class No. 3610		
Classification of Hazardous Gas & Vapor Locations	BS EN 60 079-10	ANSI/API RP 500 ANSI/API RP 505			
Classification of Mixtures of Gases & Vapors	BS 5345 Part 1 BS 5501 Part 1 BS EN 50 014 BS/EN 60079-12				
Construction of room & bldg. protected by pressurization	BS 5345 Part 5		Class No. 3620		
Area Classification in Hazardous Dust Locations	BS 6467 Part 2				ISA-S12.10
Electrical Instruments in Hazardous Dust Locations	BS 6467 Part 1		Class No. 3611		ISA-S12.11
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Country	U.S.A	U.S.A.	Croatia	P.R. of China
Standards	**NFPA**	**UL**	**DZNM/S-Commission**	
General Requirements		UL: 508, 698, 1092, 2279 Part 0	HRN N.S8 011 HRN-EN 50 014	GB3836.1
Oil Immersion/Type ‘o’			HRN N.S8 501 HRN-EN 50 015	GB3836.6
Pressurization/Type ‘p’		UL 2279 Part 2	HRN N.S8 601 HRN N.S8 602 HRN-EN 50 016	GB3836.5
Powder Filling/Type ‘q’			HRN N.S8 701 HRN-EN 50 017	GB3836.7
Explosionproof/Flameproof/Type ‘d’		UL: 674, 698, 781, 783, 823, 844, 877, 886, 894, 1002, 1010,1203,1207,1604,2225, 2279 Part 1	HRN N.S8 101 HRN-EN 50 018	GB3836.2
Increased Safety/Type ‘e’		UL 2279 Part 7	HRN N.S8 201 HRN-EN 50 019	GB3836.3
Intrinsic Safety/Type ‘i’		UL: 913, 2279 Part 11	HRN N.S8 301 HRN-EN 50 020	GB3836.4
Type ‘n’		UL: 1604, 2279 Part 15	HRN IEC 60079-15	GB3836.8
Encapsulation/Type ‘m’		UL 2279 Part 18	HRN-EN 50 028	GB3836.9
Intrinsically Safe Electrical Systems		UL 913	HRN-EN 50 039	
Classification of Hazardous Gas & Vapor Locations	NFPA 497		HRN N.S8 007 HRN-EN 60079-10	
Classification of Mixtures of Gases & Vapors	NFPA 497	UL: 58, 58A, 58B	HRN N.S8 003 HRN IEC 60079-12	GB3836.12
Construction of room & bldg. protected by pressurization	NFPA 496		HRN N.S8 610 HRN N.S8 611 HRN IEC 60079-16	
Area Classification in Hazardous Dust Locations	NFPA 499		HRN N.S8 008 HRN IEC 1241-3	
Electrical Instruments in Hazardous Dust Locations		UL: 674, 698, 781, 783, 823, 844, 877, 886, 894, 1002, 1010, 1203, 1604, 2225	HRN N.S8 850 HRN IEC 1241-1	GB12476.1
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Annex D — Listing of worldwide installation requirements

NOTE — Annex D has been filled as much as possible at the time of publication, but should not be considered complete. Additions known to the users for consideration for future editions should be directed to:

ISA
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standards@isa.org

	Australia	Austria	Belgium	China	Canada	Denmark	Finland	France	Germany	Greece	Italy	Japan
Code of Practice for selection, installation and maintenance basic requirements	AS 3000	OVE EX 65	NBN C23-201 EN 50 039 RGIE RGIE Article 251	Electrical Safety Regulations for Explosive Atmospheres of Peoples Republic of China	CEC	Heavy Current Regulations	Electrical Safety Regulations	NF C23-539 EN 50 039	EX-RL Guidelines No. 11 Elexv ZHI/ 227 Elexv ZHI/ 200 Elexv ZHI/ 309 VDE 0107 VDE 0166	PCC Regulations for interior electrical installation - Chapter X	Presiden-tial decree DPR 546 Chapter X CEI 64-2	JISHA - Recommended Practice for Explosion-Protected Electrical Installations in General Industries. JISHA - Recommended Practice for Electrical Equipment for use in Explosive Dust Atmospheres in General Industries.
Inspection & Maintenance												
Repair and Overhaul												
Selection of Instrumentation, Repair, and Maintenance for Dust Atmospheres												
Combustible Gas Detectors												

	Netherlands	Norway	Portugal	S. Africa	Spain	Sweden	Switzerland	U.K.	U.S.A.	IEC
Code of Practice for selection, installation and maintenance basic requirements	NEN 1010	NVE Communications No. 1/77	Decree No. 740/74 Decree No. 517/80 Decree No. 36270	SABS 086 SABS 0119	Instruccion MI.BT 026	SIEV-FS Safety Regulations SIND-FS Classification Regulations	SEV 1000 SEV 3538	BS/EN 60079-14	NEC/NFPA 70 API RP 14F API RP 14FZ (draft) API RP 540	IEC 60079-14
Inspection & Maintenance								BS 5345 BS/EN 60079-17		IEC 60079-17
Repair and Overhaul								BS/EN 60079-19		IEC 60079-19
Selection of Instrumentation Repair and Maintenance for Dust Atmospheres								BS 6467 Part 2 BS/EN 61241-1-2		IEC 61241-1-2
Combustible Gas Detectors								BS 6959 BS/EN 60079-14	ISA RP 12.13 Part II API RP 500 API RP 505	IEC 60079-14

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